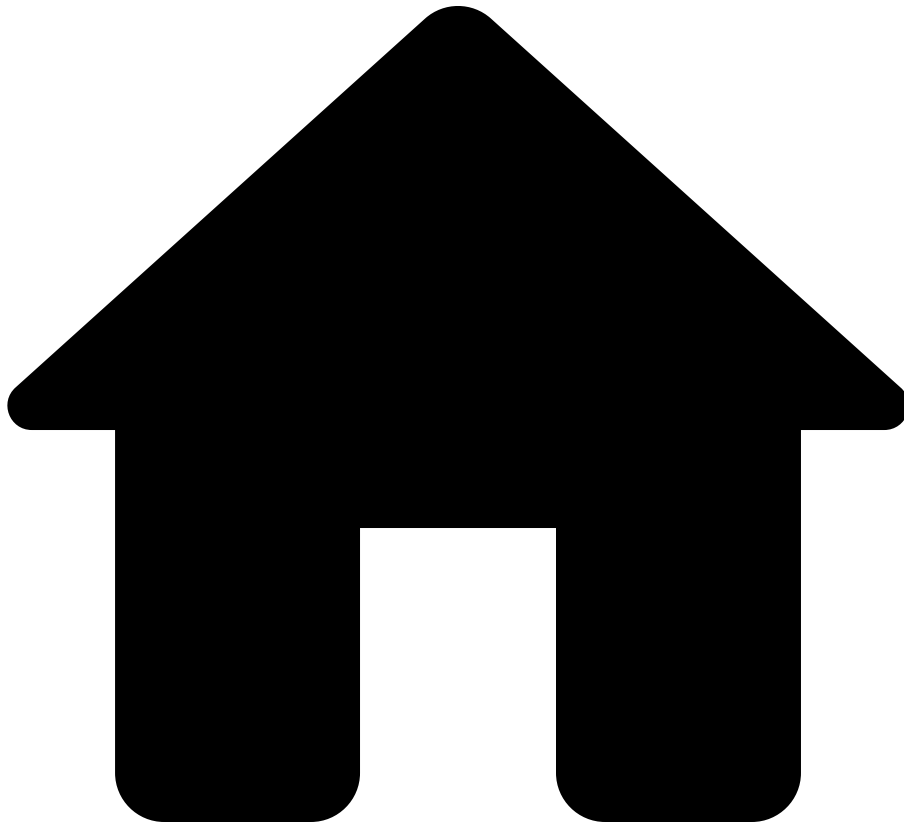


.



- [Verify](#)
- NQE Verifications

On this page

## NQE Verifications

### Overview[ââ](#)

NQE stands for **Network Query Engine**. NQE enables you to write **queries** for information about your network. The queries return a collection of records that are displayed as rows in a table.

An **NQE Verification** is formulated as a query for violations. Every NQE Verification follows a common pattern: query the data model to identify violations of the property that you want to verify. If the query finds no violations, then the verification passes. If the query finds violations, the verification is marked as failed, and the identified violations form the failure diagnosis. You can create, store, run, diagnose and publish your **NQE queries** entirely in the Forward Enterprise [NQE Library](#) application. The NQE queries need to be published in the NQE Library application via the **Commit** process in order to be used as NQE Verifications.

## Getting started

Follow instructions in the editor to get started writing your first check. Some key points to notice:

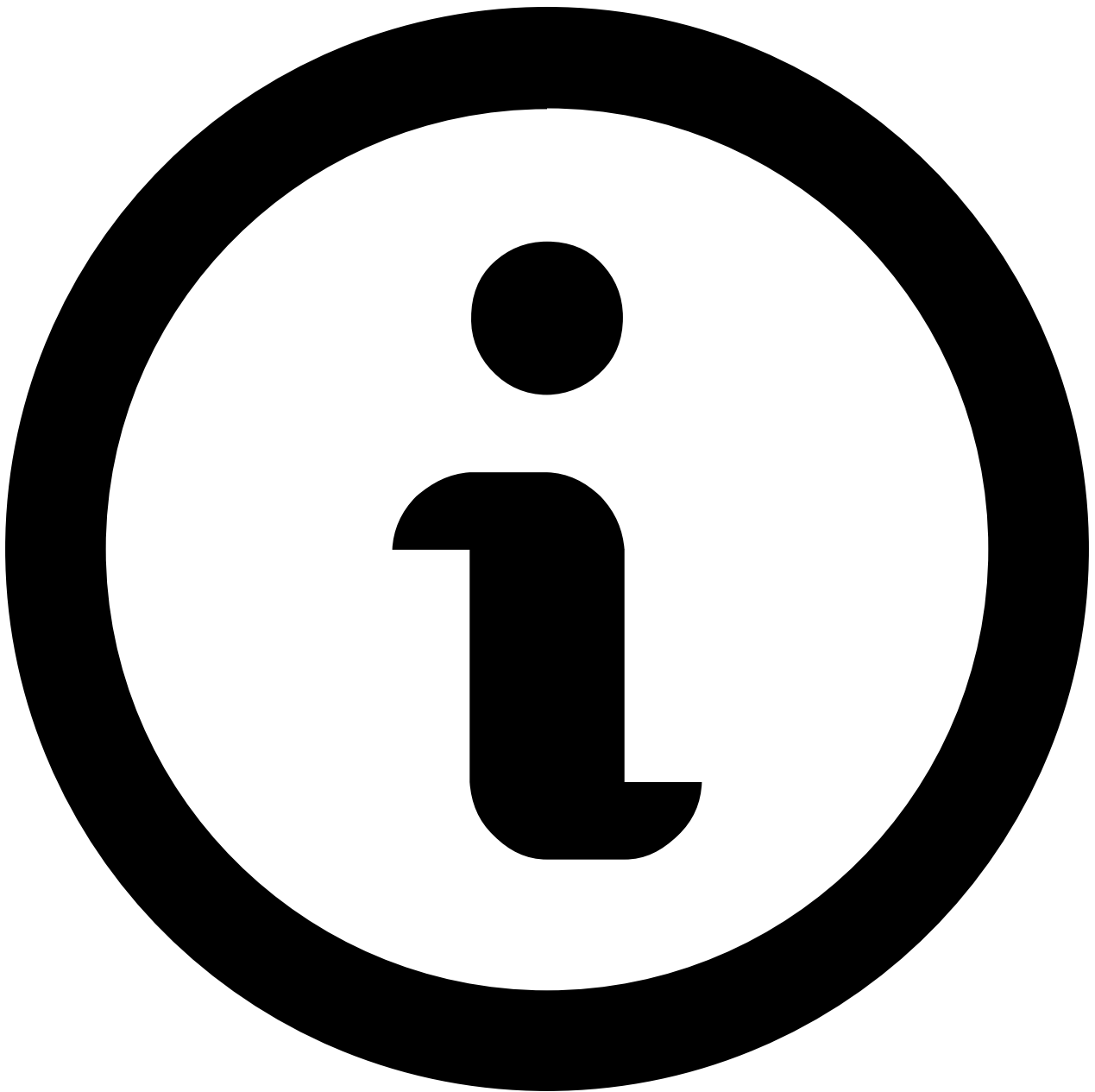
- Results show up below in the *Results* tab.
- If there are problems with your edits, the editor will underline problematic parts. Hover over the underlined part to see a detailed description and suggestion for how you might fix the problem.
- The [Overview section](#) describes the key parts of an NQE Check.

## NQE Checks with `violation` label

An **NQE Check** is formulated as an NQE query. Each row may include the label `violation` containing a value of type `Bool` indicating whether the row is a violation or a passing verification. The rows of an NQE Check should include all attributes of the objects of interest that are relevant to understanding why the row is labeled as a violation or a passing verification. This allows readers of the check results to quickly understand what the NQE check is verifying and why an item may be failing.

If any rows returned by an NQE query are labeled as violations, then the check is marked failed and the attributes in each violation row forms the diagnosis for the failure indicated by the row. Otherwise, the check passes, and the rows can be used to inspect what network items were positively verified.

The numbers of passing and failing rows are charted on the Dashboard as the number of passing verifications and the number of violations, respectively.



note

The letters in the `violation` label must be all lowercase. If some letter is uppercase, then a warning will be given.

### **NQE Checks with no `violation` label**[↗](#)

If an NQE Check does not include a `violation` label to indicate whether each row is violation or passing verification, it is assumed that every row is a violation. In addition, for such an NQE Check, if the query produces no rows, then the number of passing

verifications is 1 and the number of violations is 0. Otherwise, the number of passing verifications is 0 and the number of violations is 1.

Note that this type of check is deprecated and allowed for backwards-compatibility. Support for this type of check will be removed in a future release.

## Anatomy of a Check

NQE Checks use a syntax similar to [LINQ](#), which is itself similar to SQL.

For example, the following check finds every interface on Cisco NXOS devices. It marks each interface that is admin UP, but not operationally UP as a violation. All other interfaces are considered passes.

```
foreach device in network.devices
foreach interface in device.interfaces
where device.platform.os == OS.NXOS
select {
  deviceName: device.name,
  interfaceName: interface.name,
  adminStatus: interface.adminStatus,
  operStatus: interface.operStatus,
  violation: interface.adminStatus == AdminStatus.UP &&
             interface.operStatus != OperStatus.UP
}
```

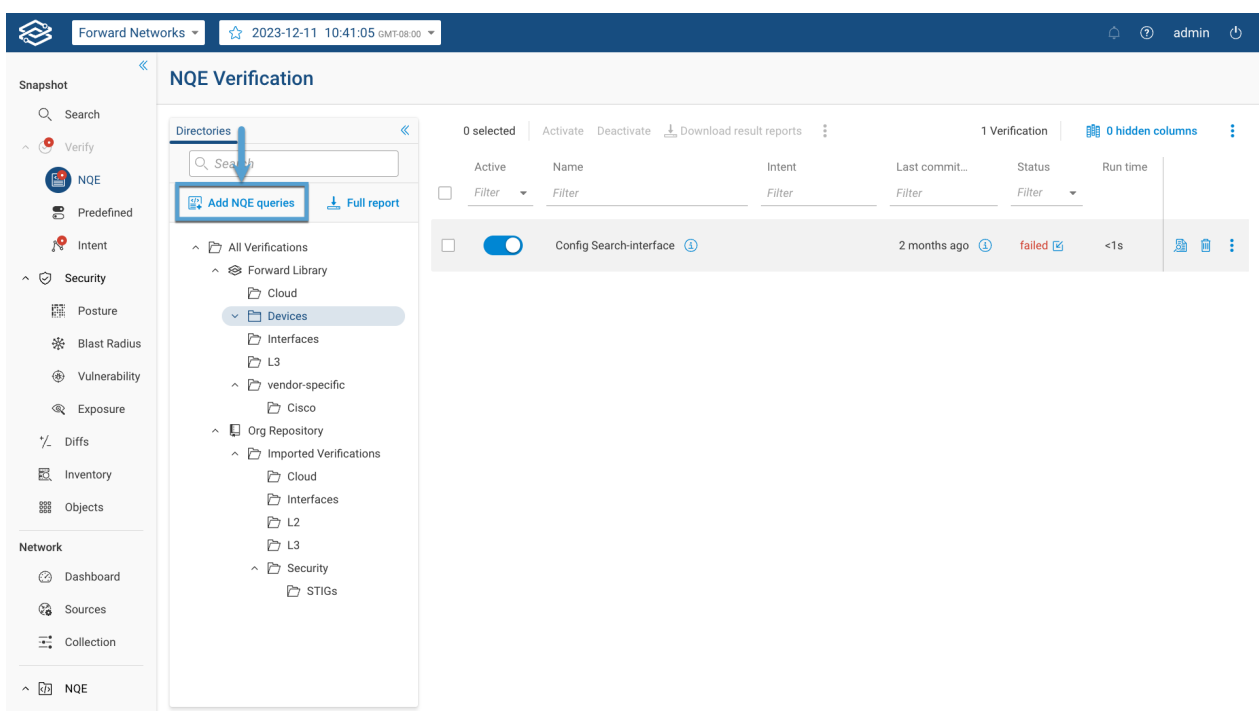
The basic idea is that `foreach` clauses iterate over the data, the `where` clauses restrict (filter) the data to the condition you care about, and the `select` clause determines what information is included in the diagnosis. Expanding on this:

- **foreach** : In the above check, the first two `foreach` clauses iterate over the devices and their interfaces. `foreach` only applies to collections of elements. There can be any number of `foreach` clauses.
- **Scoping**: `foreach` introduces a new variable that is available in the subsequent lines of the check. In the above example, `device` and `interface` are introduced in the first two lines and then used in the last few lines of the check.
- The only variable in scope initially is `network`. This variable provides access to all attributes of the network.

- **where** : The **where** clause limits the data to just that matching the condition you care about. Think of it as a way to filter data. The **where** clause supports any boolean-valued condition, including expressions constructed from familiar boolean and relational operators, such as **&&** (and), **||** (or), **!** (not), **==** , **!=** , **<** , **>** , etc. You can have any number of **where** clauses. They can appear intermixed with **foreach** clauses, so long as they only use variables in scope at that line.
- **select** : The final **select** clause constructs a record (or object) for each combination of **device** and **interface** values that satisfy the **where** condition. These records determine what information is included in the diagnosis. The field names are used as column names in the results table. There can be only one **select** clause.
- **violation** field: Each row produced in the **select** clause should have a field called **violation** with a boolean-valued condition. This condition defines whether a row is considered a violation or a passing verification.
- A record is a comma-separated list of name-value pairs enclosed in curly braces: `{name1: value1, ...}`.
- Case-sensitivity: The language is case-sensitive.
- Whitespace: The language is not sensitive to the number of spaces or line breaks.

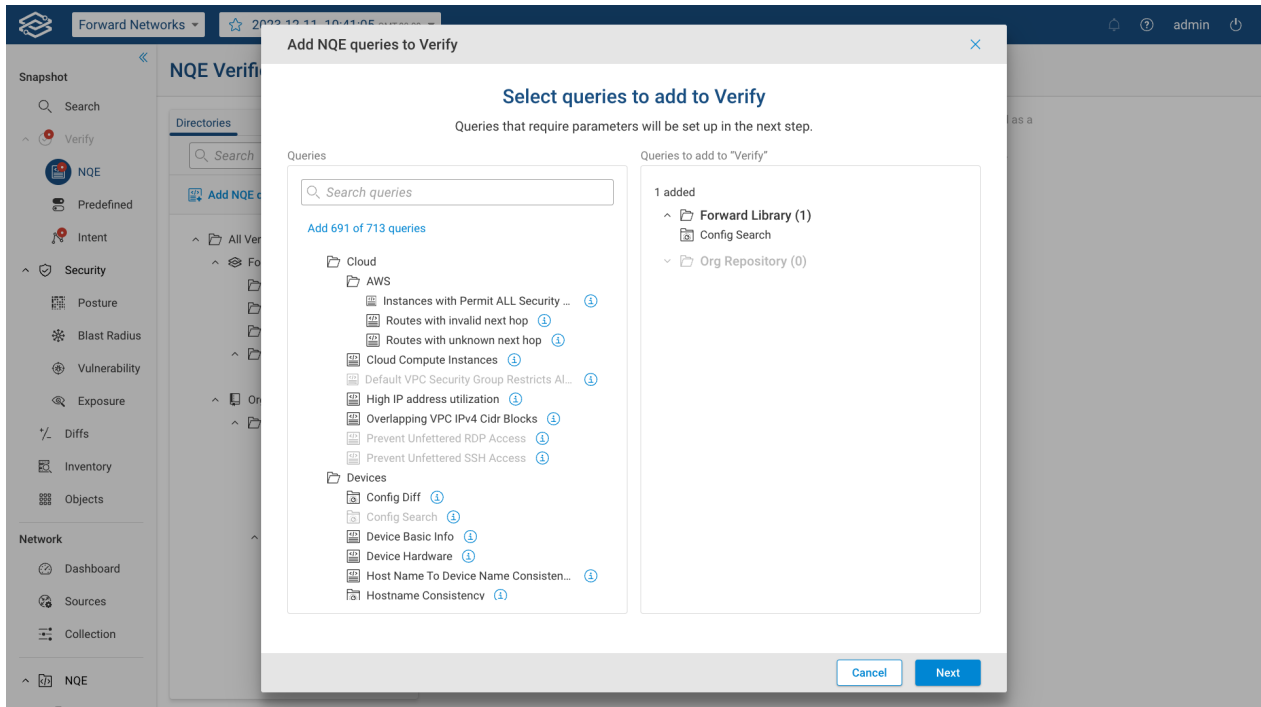
## Adding NQE Queries to NQE Verifications

To add NQE queries to NQE Verifications, initiate the process by selecting the **Add NQE Queries** button within the NQE Directory Panel.



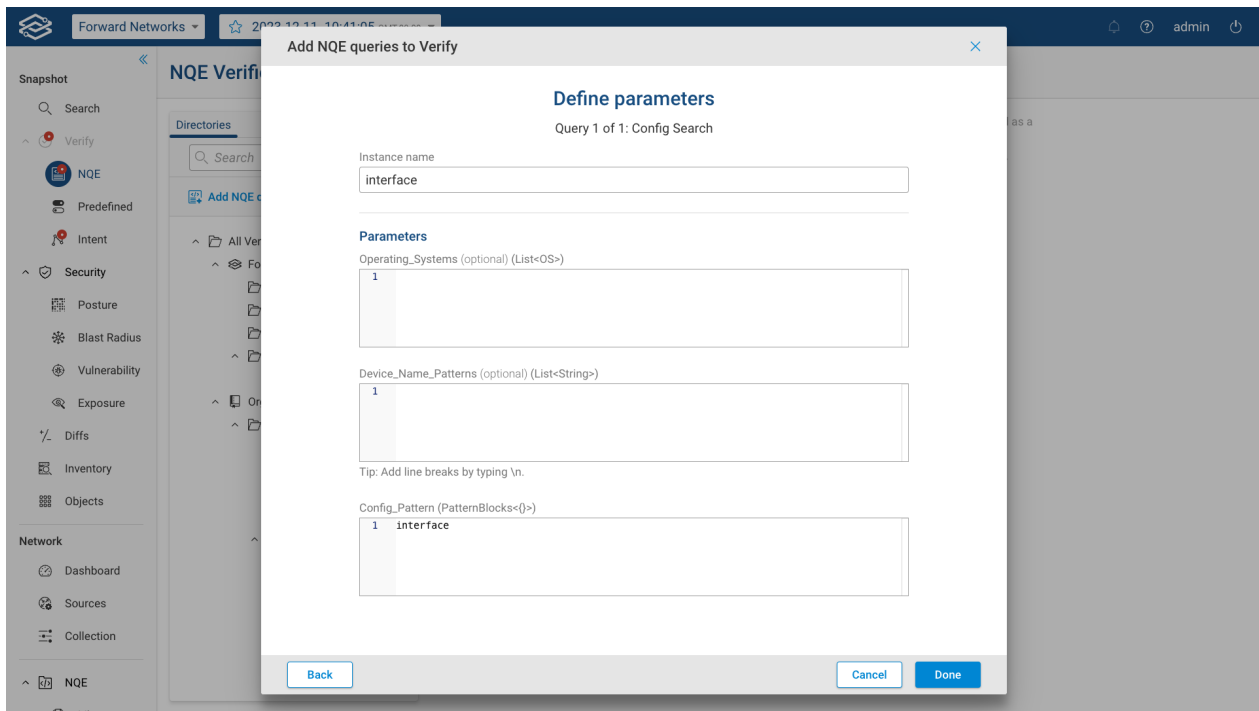
## Selecting NQE Queries

Choose one or more queries from the available directories to add as NQE Verifications. The selected queries will be listed in the adjacent panel.



## Defining Query Parameters

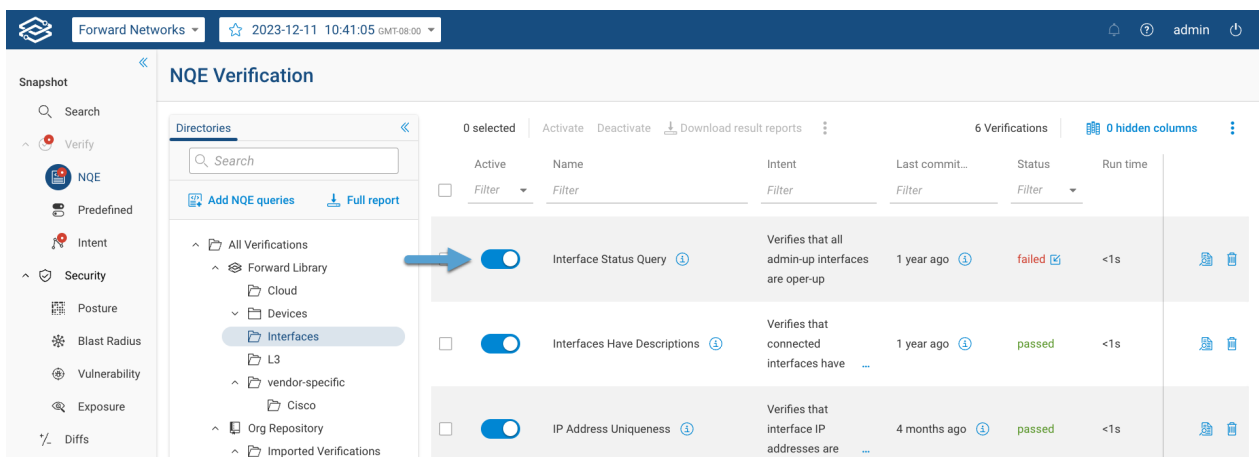
For NQE queries requiring additional parameters to be configured, define them before adding. For each query, provide a Name and enter the preferred parameter values. This process repeats until all NQE query parameters are assigned.



Visit the [Parameterized Queries](#) section of the [NQE Library](#) page for additional information on parameterized queries.

## Enabling/Disabling NQE Verifications

By default, newly added NQE queries start in an enabled (**active**) state. Clicking on the **Active** toggle of an NQE Verification will change its state to disabled (**inactive**).



## Deleting NQE Verifications

To delete an NQE Verification, select the **Delete** icon, located at the end of each row, to delete it.

# NQE Verification Results

If an NQE Verification fails, the violations and query are available by clicking on the **Failed** status link, or by selecting the **View Details** icon. The **Results** tab displays the report, while the Code tab displays the NQE query used to generate the report.

Forward Networks

2023-12-11 10:41:05 GMT-08:00

admin

Snapshot

Search

Verify

NQE

Predefined

Intent

Security

Posture

Blast Radius

Vulnerability

Exposure

Diffs

Inventory

Objects

Network

Dashboard

Sources

Collection

NQE

Interface Status Query

ResultsCode

112 of 112 results

Download XLSX report

Download CSV report

| violation | deviceName                                  | interfaceName                               | adminStatus | operStatus |
|-----------|---------------------------------------------|---------------------------------------------|-------------|------------|
| Filter    | Filter                                      | Filter                                      | Filter      | Filter     |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | vgw-2f2ccd46/dxvif-fg7zr93m.1               | UP          | UP         |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | bdr-102-2.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | vgw-34f41f5d/dxvif-ffyg5gw3.2               | UP          | UP         |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | bdr-102-2.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | vgw-2f2ccd46/dxvif-fh7abxm0.1               | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | bdr-103-1.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | vgw-34f41f5d/dxvif-ffygcs70.2               | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | bdr-103-1.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| true      | MPLS                                        | LAX-MPLS-RT01/gi0/0                         | UP          | DOWN       |

Close

# Previewing Results with Updated Data Files

Some NQE-based verifications depend on external data files. When a data file is updated, the verification's details view includes a selector that allows you to run the check using:

- **Snapshot** the authoritative snapshot-based result
- **Snapshot with Latest Data Files** a preview that overlays the most recently uploaded data files onto the snapshot

This allows you to validate updates to data files immediately, without waiting for a new snapshot collection. The main verification status remains based on the snapshot.



Software Compliance

ResultsCode

Data filesFrom SnapshotLatest

32 of 32 filtered resultsDownload report

DeviceOSOS VersionCompliance StatusviolationTarget Versions

FilterFilterFilterFilterPass (17), ...Filter

PHX-BLDG7-DIST-1.net3.orgARISTA\_EOS4.31.2FTARGETPass4.31.2F

PHX-BLDG7-DIST-2.net3.orgARISTA\_EOS4.31.2FTARGETPass4.31.2F

SJ-BLDG1-C9800-WLC10IOS\_XE17.13.01ACCEPTABLEPass17.12

SJ-BLDG1-CM-CORE-1ARISTA\_EOS4.31.2FTARGETPass4.31.2F

SJ-BLDG1-SP-EDGE-01IOS\_XE17.09.01aACCEPTABLEPass17.12

SJ-BLDG1-SP-EDGE-02IOS\_XE17.09.01aACCEPTABLEPass17.12

Close

## STIG Compliance

Forward includes a built-in **STIG Findings** verification in the Forward Library, allowing you to assess compliance against supported DISA STIG rules.

In addition to the main STIG Findings check, the Library also provides **severity-specific STIG Findings queries** (for example, queries scoped to Critical, High, or Medium severity findings).

These checks can be add from Forward Library directly to the Verify page and behave like other NQE-powered verifications.

NQE Verification

Directories

Add NQE queries

Full report

All Verifications

Forward Library

Cloud

Interfaces

L3

Security

STIGs

vendor-specific

Org Repository

0 selected

Activate

Deactivate

Remove

Recompute

4 Verifications

1 hidden column

| Active                   | Name                    | Intent                                                                   | Verification status | Priority |
|--------------------------|-------------------------|--------------------------------------------------------------------------|---------------------|----------|
| Filter                   | Filter                  | Filter                                                                   | Filter              | Filter   |
| <input type="checkbox"/> | STIG Findings           | Verify compliance with all Forward-supported STIG rules.                 | failed              | medi...  |
| <input type="checkbox"/> | STIG Findings - Cat I   | Verify compliance with all high severity Forward-supported STIG rules.   | failed              | high     |
| <input type="checkbox"/> | STIG Findings - Cat II  | Verify compliance with all medium severity Forward-supported STIG rules. | failed              | medi...  |
| <input type="checkbox"/> | STIG Findings - Cat III | Verify compliance with all low severity Forward-supported STIG rules.    | failed              | low      |

## STIG Security Policy (User-Defined Parameters)

Some STIG rules require operator-supplied parameters to determine whether certain conditions apply to a device or interface.

These parameters are provided through the **STIG Security Policy** file ( `fn-stig-policy.csv` ), imported as a data file.

You can download the policy template from:

**NQE Data Files Import data file Download Template**

Import data file

Import data file

Upload your data file to begin the import process. Make sure that:

- .json or .csv extensions are in the file name
- STIGs policy file must follow a specific format. Download and fill it out the provided template before uploading it.

Download template

Note: Only one STIG policy file can exist in the system at any given time. Uploading a new policy file will replace an older version.

Drag and drop file here

The import process will start immediately after you select the file.

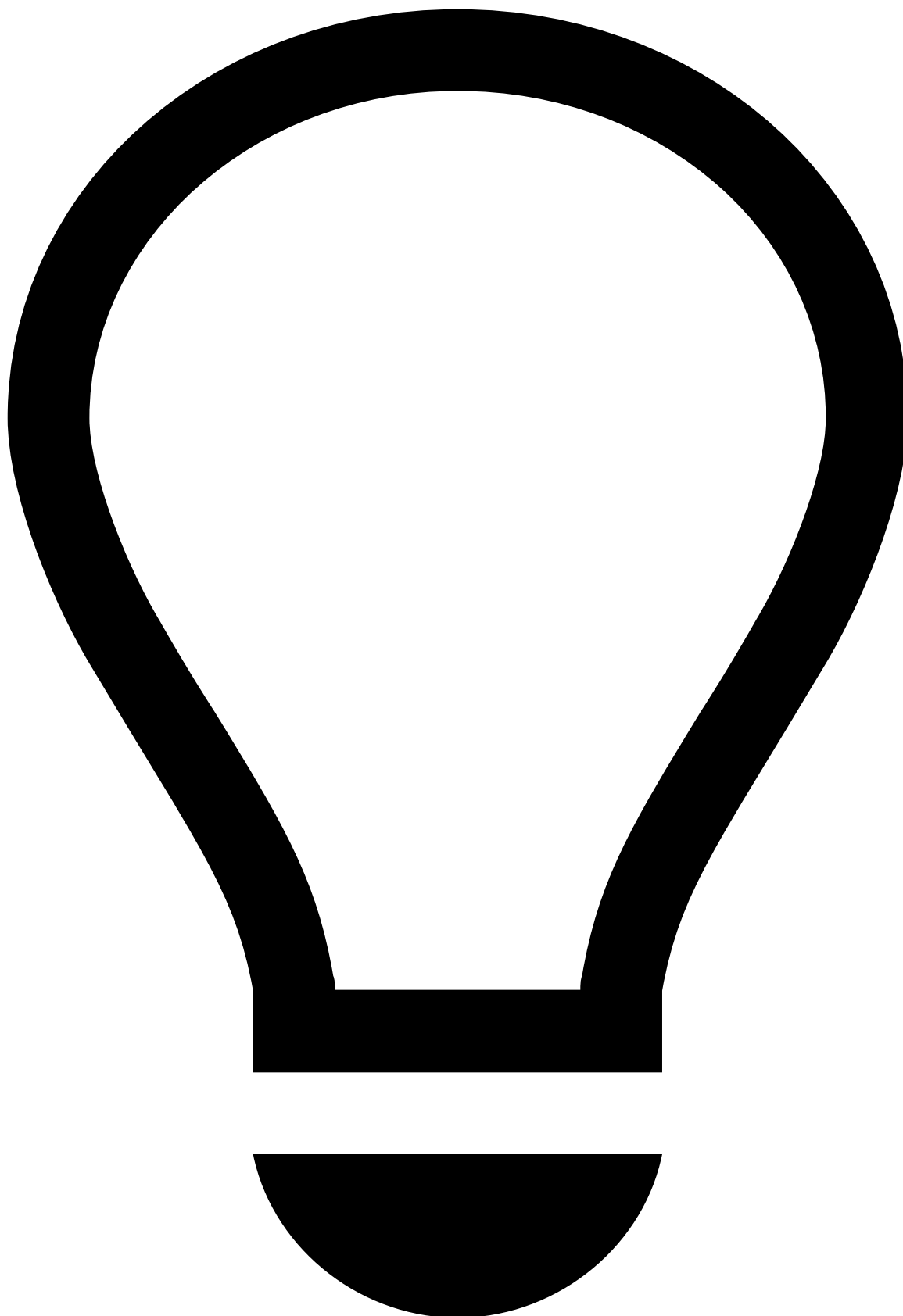
or

Select file

Cancel

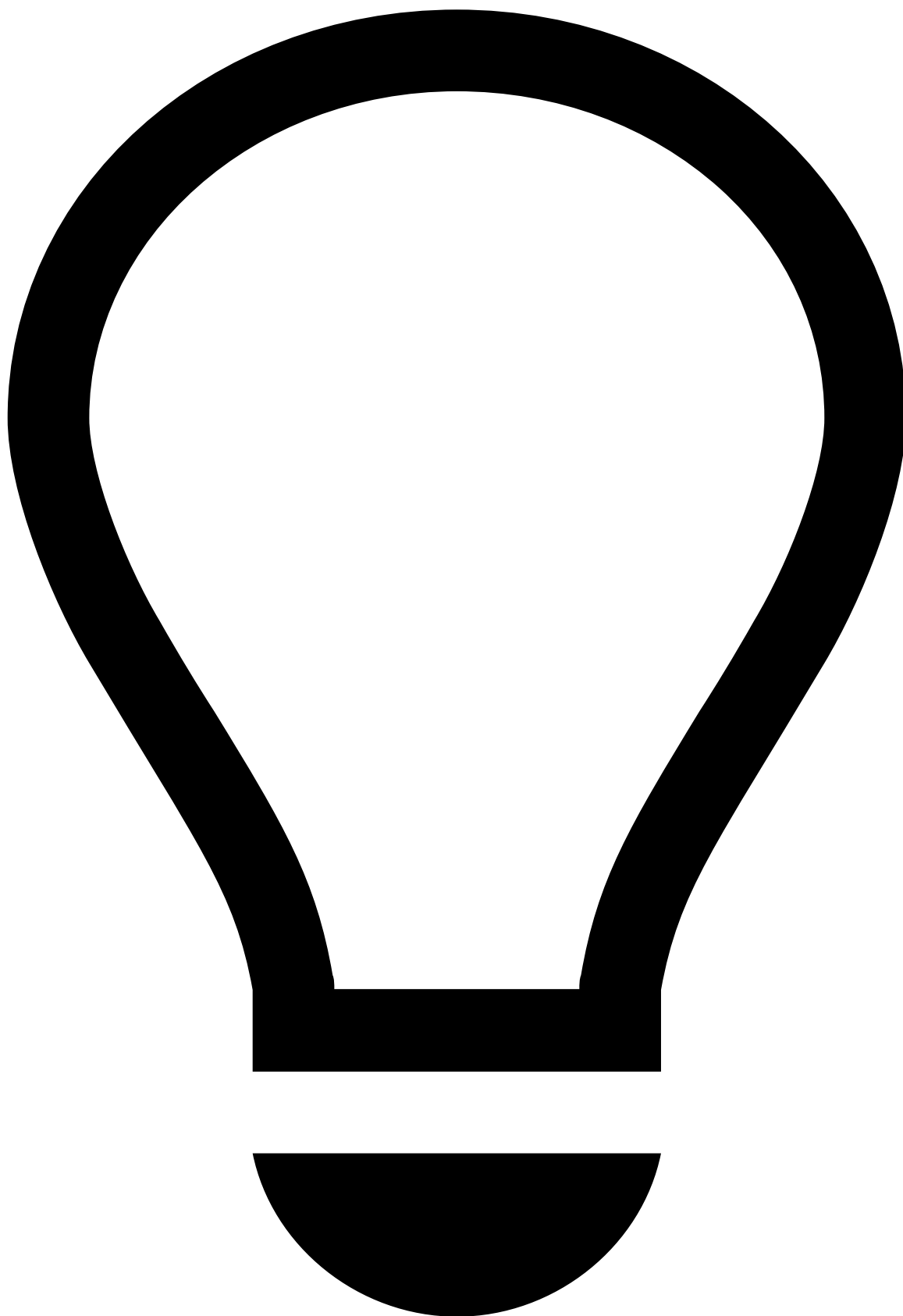
Next

After editing the policy values, upload the file using **NQE &#x2013; Data Files**. Once uploaded and enabled for the network, the STIG Findings verification automatically incorporates these parameters.



tip

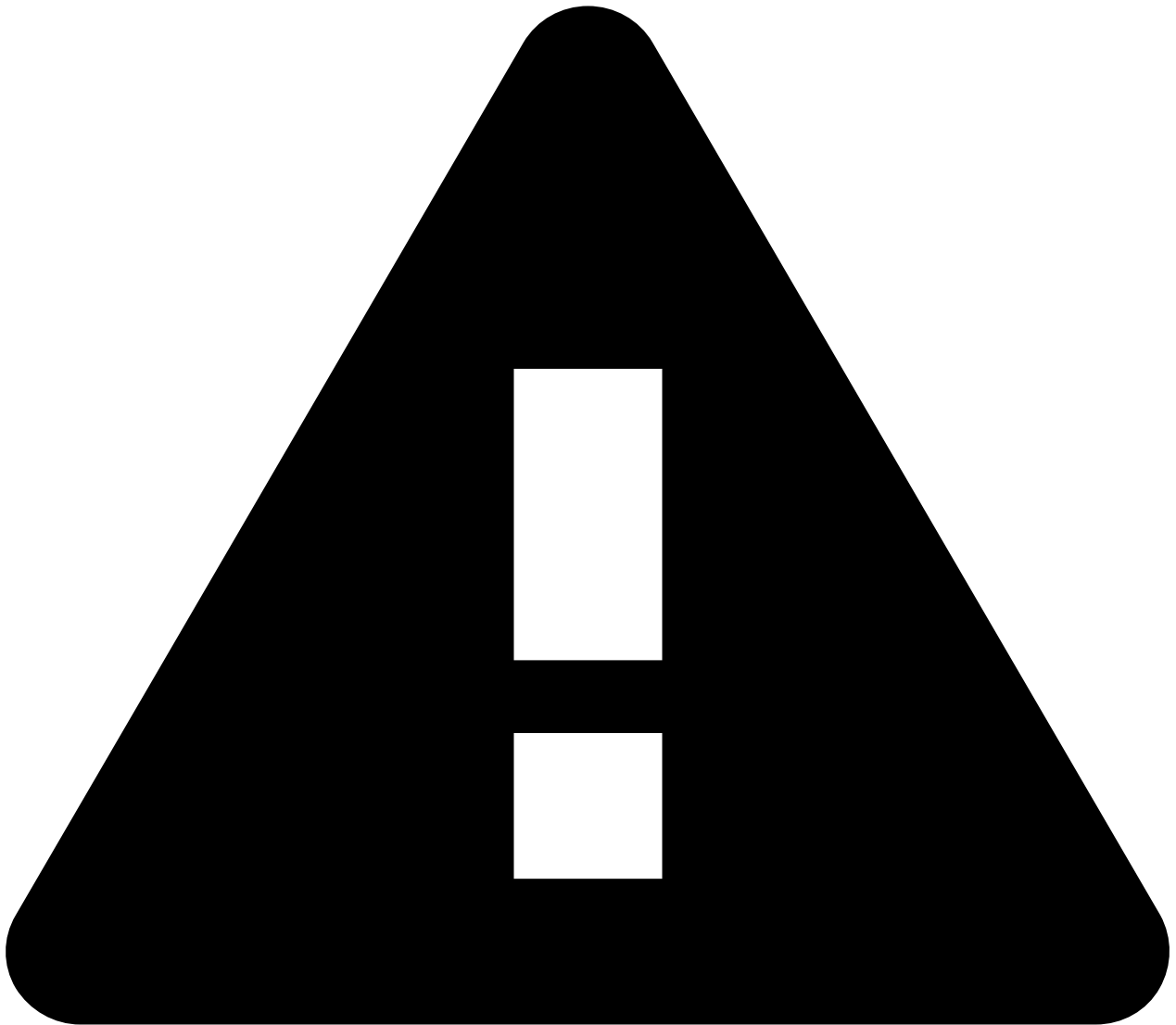
Visit the [Data Files](#) page for more details on managing data files.



tip

STIG Findings and severity-specific STIG queries can be run using the latest uploaded STIG Security Policy file, without waiting for the next snapshot. Use the selector in the verification details view to choose Snapshot or Snapshot with Latest Data Files.

See [Data Files for STIG Security Policies](#) for more details



[Important]

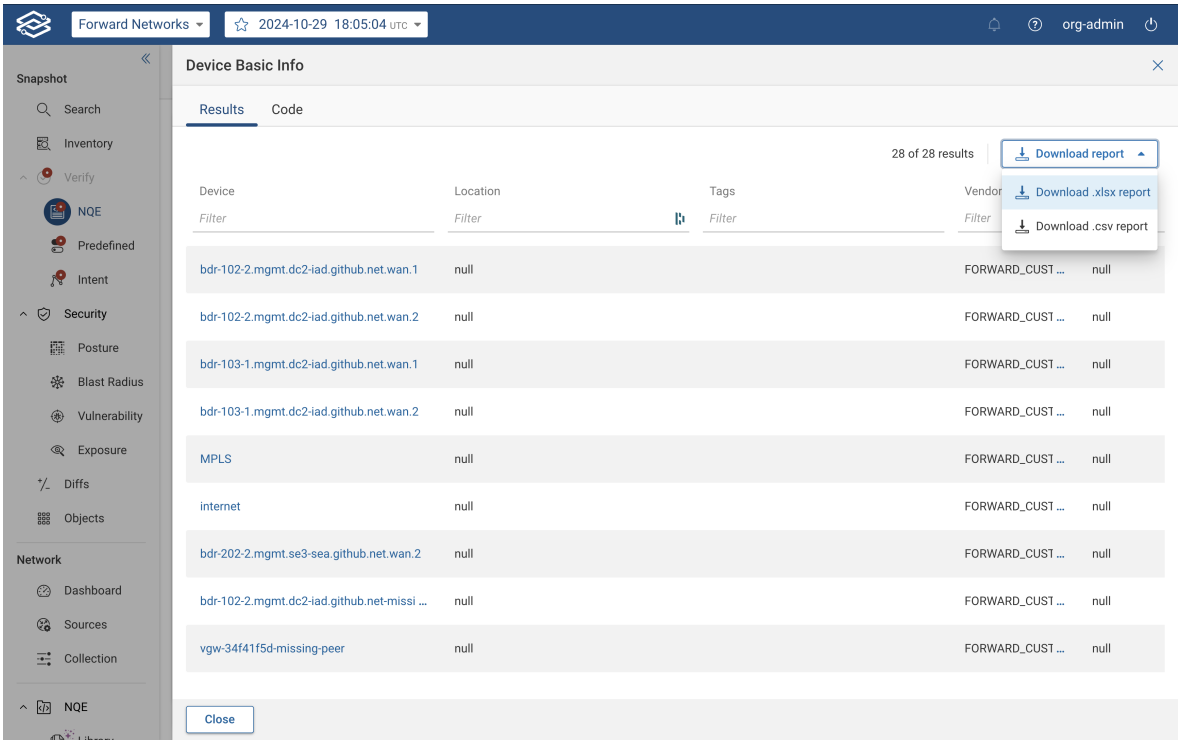
The verification status shown in the main Verify table always reflects the snapshot only. Previewing with latest data files does not modify snapshot-driven results.

# Downloading a Report🔗🔗

The **NQE Verifications** page allows users to export query results in various formats for analysis or compliance purposes. Available export formats include **CSV**, **XLSX**, and **CKLB**.

## Downloading a CSV and XLSX Report🔗🔗

1. Open the **NQE Verifications** page and locate the desired query and click the **View Details** icon.
2. In the results pane, click on **Download report** at the top-right corner.
3. Select one of the following options from the dropdown:
  - **Download .xlsx report**
  - **Download .csv report**



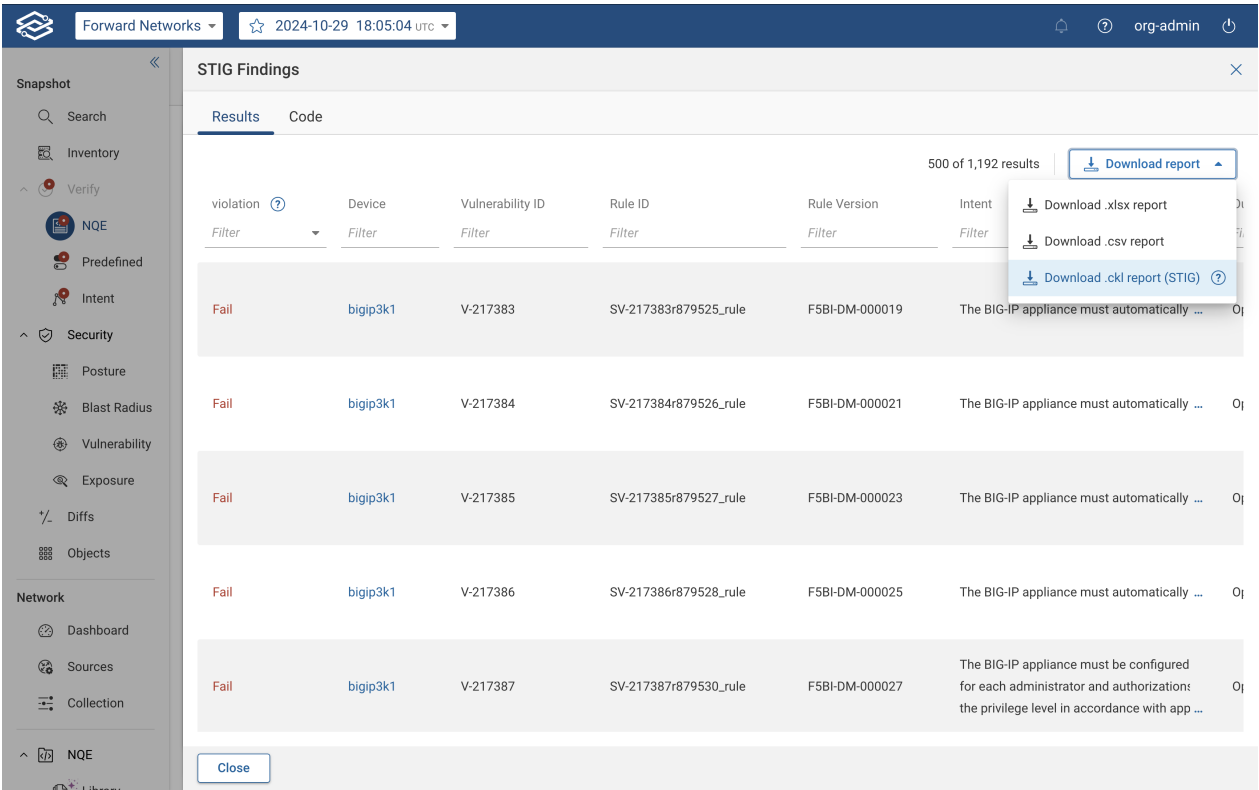
## CKLB Export🔗🔗

CKLB Export generates STIG compliance reports in a DoD-compliant [CKLB format](#). This export option is available only for queries that include all the required columns found in any of the **STIG Findings** queries (e.g., STIG Findings - Cat I, STIG Findings - Cat II, or

STIG Findings - Cat III). While additional columns can be added to the query, none of the required columns can be removed.

To download a STIG report:

- 1. Open the **NQE Verifications** page and locate any of the **STIG Findings** queries.
- 2. In the results pane, click on **Download report** at the top-right corner.
- 3. Select **Download .ckl report (STIG)** from the dropdown.



## Additional Options

An **Additional Options** icon is available for Verifications with defined query parameters:

- **Edit Parameters** - Selecting this option opens the edit parameter settings window for editing.
- **Duplicate** - Selecting this option will duplicate the selected Verification and make a copy available for use.

Forward Networks

2023-12-11 10:41:05 GMT-08:00

admin

Snapshot

Search

Verify

NQE

Predefined

Intent

Security

Posture

Blast Radius

Vulnerability

Exposure

NQE Verification

Directories

Search

Add NQE queries

Full report

All Verifications

Forward Library

Cloud

Devices

Config Search

Interfaces

L3

BGP

0 selected

Activate

Deactivate

Download result reports

2 Verifications

0 hidden columns

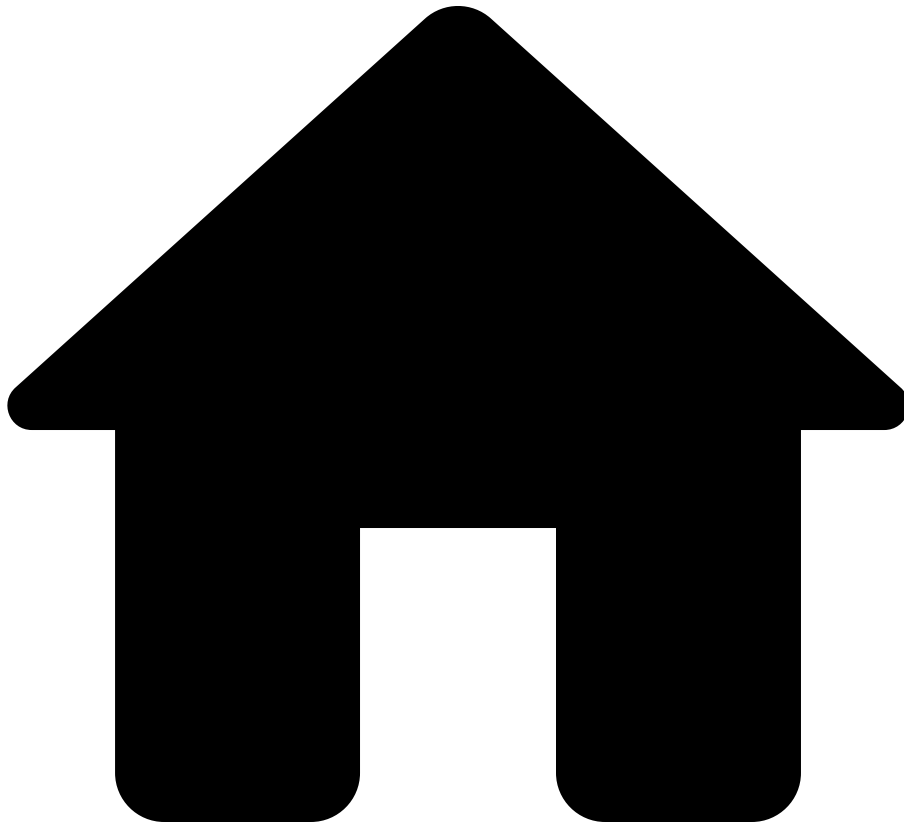
| Active                              | Name                     | Intent                   | Last committed           | Status                   | Run time                 | Priority                 |
|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Filter                              | Filter                   | Filter                   | Filter                   | Filter                   | Filter                   | Filter                   |
| <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| <input checked="" type="checkbox"/> | Config Search-interface  |                          | 2 months ago             | failed                   | <1s                      | not set                  |
| <input checked="" type="checkbox"/> | Config Search-interfaces |                          | 2 months ago             | failed                   | <1s                      | not set                  |

Edit parameters

Duplicate



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- Security
- Vulnerability Analysis

On this page

# Vulnerability Analysis

## Overview

Network device operating systems can be vulnerable to security flaws. Publicly disclosed vulnerabilities are assigned a **CVE** (Common Vulnerabilities and Exposures) **ID number** and a severity level based on their impact. CVEs help IT professionals prioritize and address these vulnerabilities to secure systems and networks.

Networking vendors often fix security vulnerabilities by issuing new OS versions. Therefore, network operators need to evaluate their devices' operating systems regularly. However, the number of vulnerabilities and affected devices can be overwhelming, making it difficult to determine which devices should be updated first.

## OS Vulnerability Analysis

To simplify this process, Forward Enterprise provides a network device vulnerability analysis tool. This tool automatically compares CVE information from the vendors and the [NIST National Vulnerability Database \(NVD\)](#) with the OS versions running on your devices, producing a list of **possibly affected devices** and related vulnerabilities.

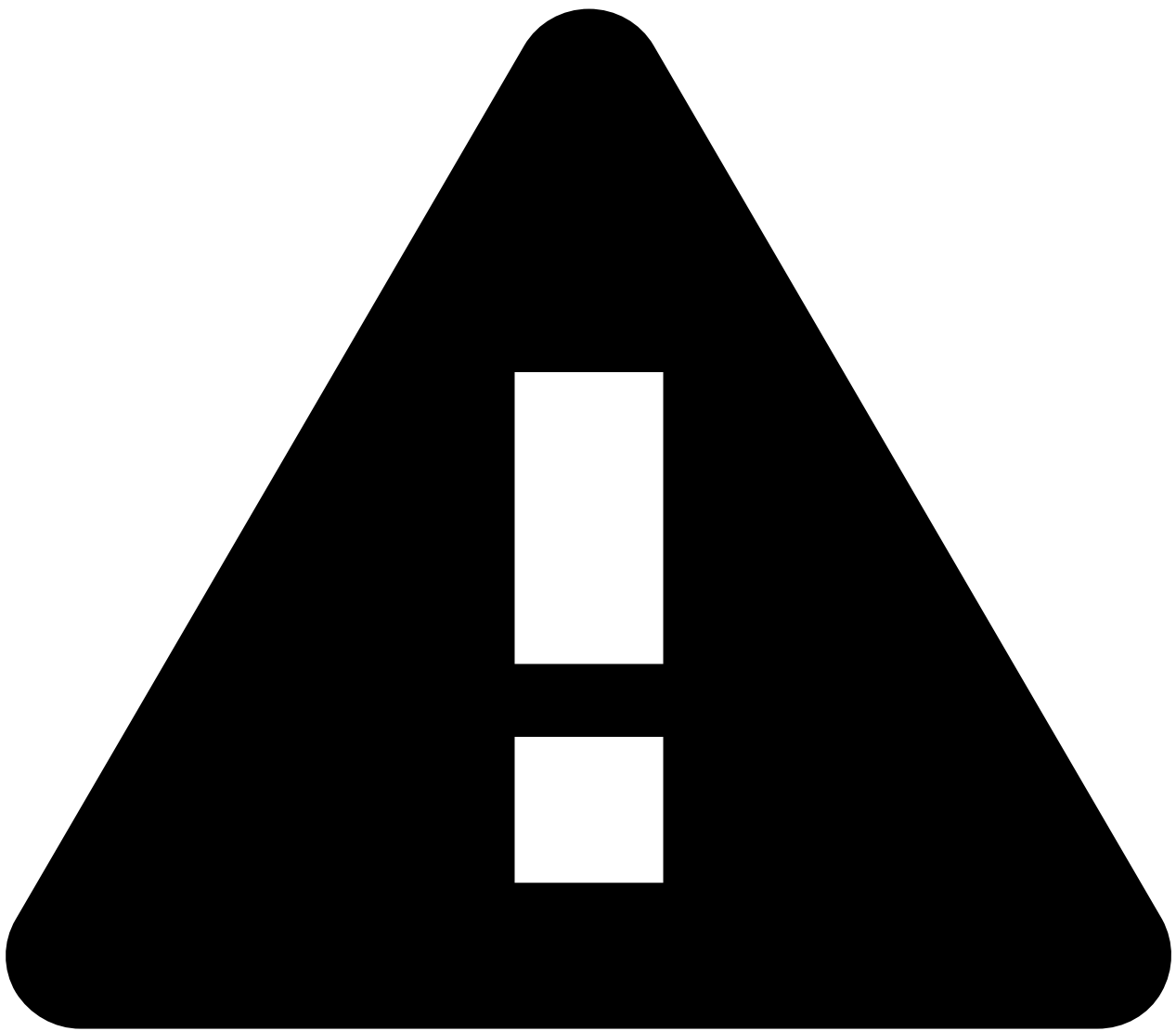
**CVE Database Updates:** The CVE database on Forward SaaS is updated daily. On-premises deployments receive the CVE database with each release but can also update it independently. For more information, see [Update CVE database](#).

## Enhanced Vulnerability Analysis

Some vulnerabilities impact devices only if specific configurations are present, specific features are enabled, or the device is deployed in a certain way.

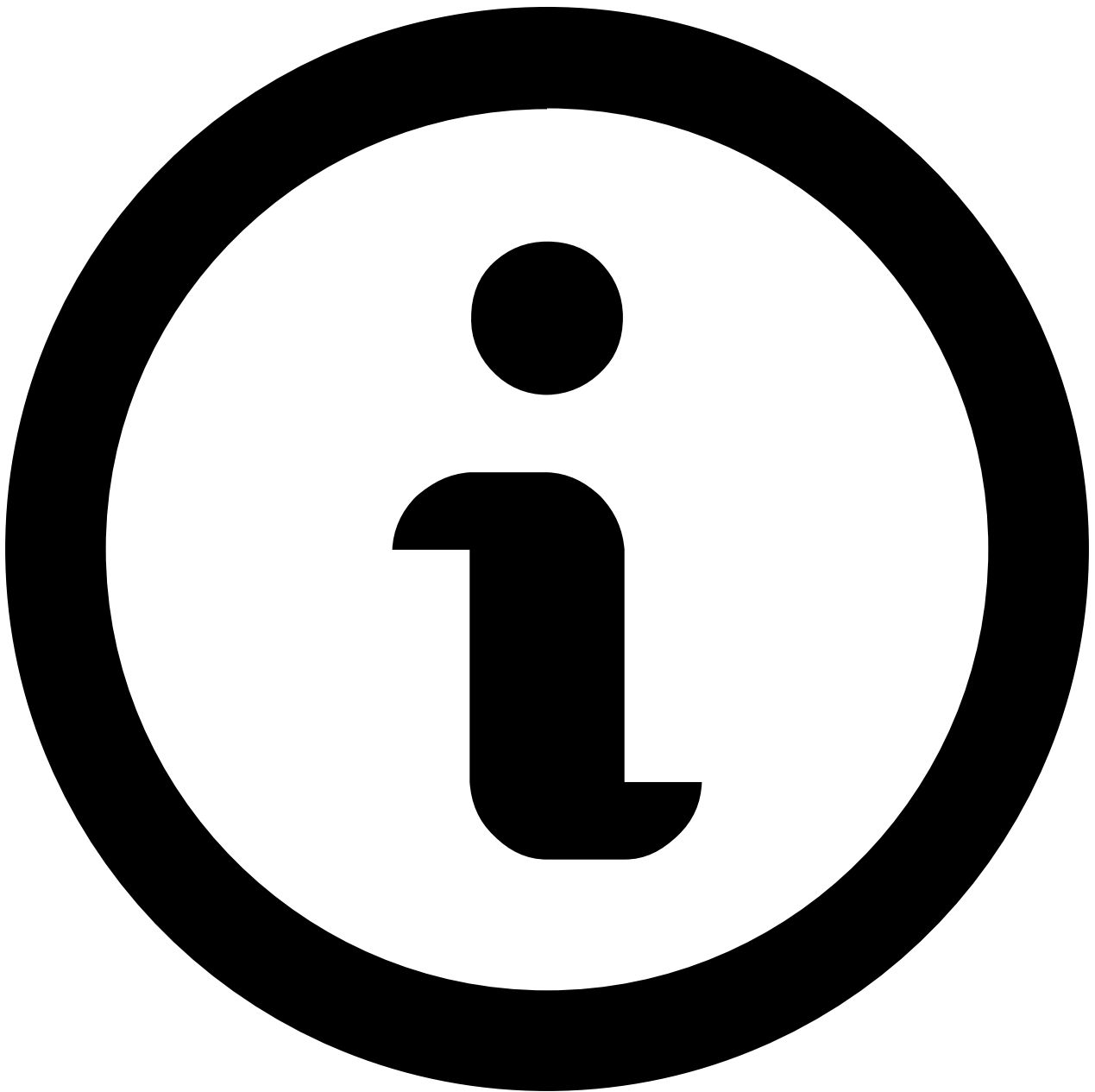
Some vendors, like Cisco, provide detailed descriptions of some CVEs, including required configurations that are vulnerable. This information is available at vendor-specific sites; for example, the **Cisco Security Advisories** can be found [here](#).

Forward Enterprise's **Enhanced Vulnerability Analysis** incorporates these descriptions and automatically checks them against device configurations. This helps determine a device's actual vulnerability status, aiding in prioritizing remediation efforts.



warning

Even with enhanced analysis, we always recommend checking the details of the CVE before deciding the next appropriate actions.



#### note

The specific list of operating system (OS) versions that include enhanced analysis is available in the Feature Matrix section of the documentation. Forward Enterprise will continue to expand the number of CVEs and vendors in future releases of the Forward vulnerability database.

The following NQE query can be used to retrieve the complete list of device vendors and OSes with enhanced analysis support:

```

foreach cveDatabase in [network.cveDatabase]
foreach cve in cveDatabase.cves
foreach vendorInfo in cve.vendorInfos
foreach criteria in vendorInfo.criteria
foreach osCriteria in criteria.osCriteria
where isPresent(osCriteria.dependsOnConfig)
select distinct {
  Vendor: vendorInfo.vendor,
  OS: osCriteria.os
}
order by Vendor asc, OS asc

```

## Navigating the Vulnerability Analysis Page

To access the Vulnerability Analysis page, navigate to **Security** > **Vulnerability** from the left navigation menu.

Each tab within the Vulnerability Analysis page presents key vulnerability metrics, along with two detailed views for CVE-specific information: [Devices](#) and [CVEs](#).

The information in both tabs is broken down in different sections as highlighted in the following picture:

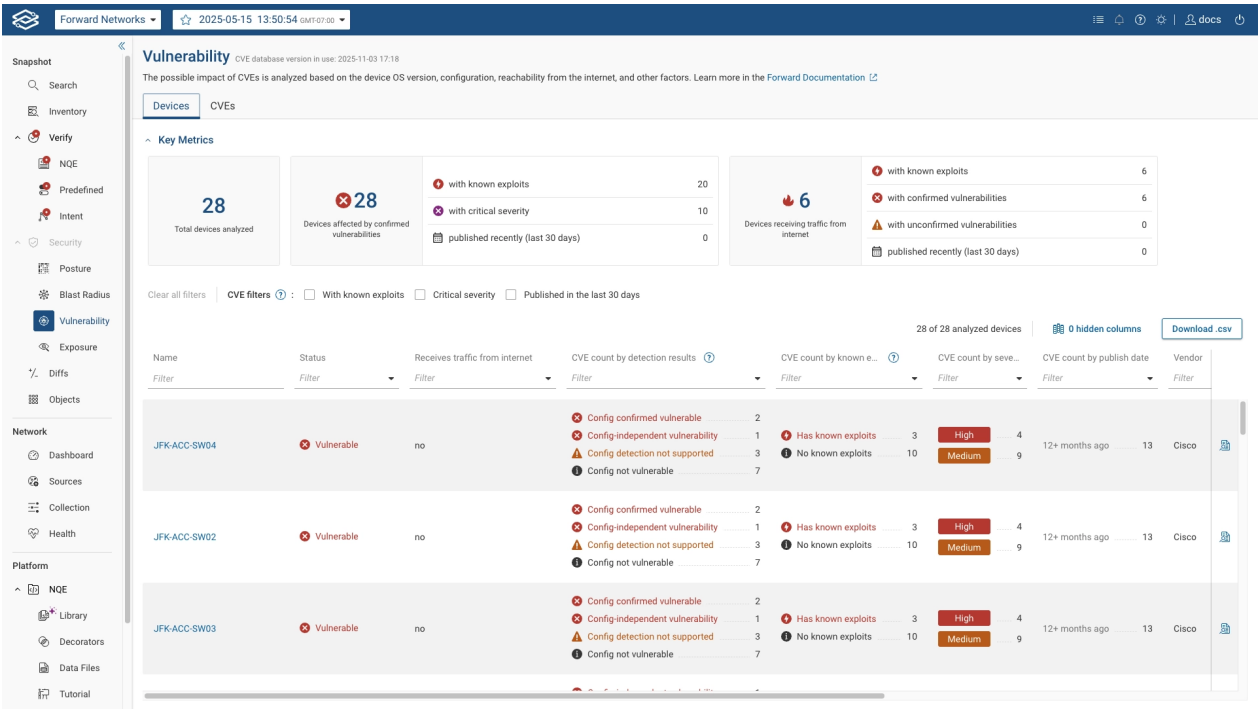
The screenshot displays the 'Vulnerability' page in the Forward Networks interface. The left sidebar shows the navigation menu with 'Vulnerability' selected. The main content area is titled 'Vulnerability' and includes a sub-tab for 'Devices'. The 'Key Metrics' section shows 28 total devices analyzed, with 28 devices affected by confirmed vulnerabilities. The 'List' section displays a table of devices, including their names, status, and CVE counts. Annotations highlight the 'Views' (Devices/CVEs tabs), 'Key Metrics' section, 'Filters' (above the list), and 'List download' button.

| Name         | Status     | Receives traffic from internet | CVE count by detection results                                                                                                                                                                             | CVE count by known e...                                                                                | CVE count by seve...                                                         | CVE count by publish date | Vendor |
|--------------|------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------|--------|
| JFK-ACC-SW04 | Vulnerable | no                             | <ul style="list-style-type: none"> <li>Config confirmed vulnerable: 2</li> <li>Config-independent vulnerability: 1</li> <li>Config detection not supported: 3</li> <li>Config not vulnerable: 7</li> </ul> | <ul style="list-style-type: none"> <li>Has known exploits: 3</li> <li>No known exploits: 10</li> </ul> | <ul style="list-style-type: none"> <li>High: 4</li> <li>Medium: 9</li> </ul> | 12+ months ago            | Cisco  |
| JFK-ACC-SW02 | Vulnerable | no                             | <ul style="list-style-type: none"> <li>Config confirmed vulnerable: 2</li> <li>Config-independent vulnerability: 1</li> <li>Config detection not supported: 3</li> <li>Config not vulnerable: 7</li> </ul> | <ul style="list-style-type: none"> <li>Has known exploits: 3</li> <li>No known exploits: 10</li> </ul> | <ul style="list-style-type: none"> <li>High: 4</li> <li>Medium: 9</li> </ul> | 12+ months ago            | Cisco  |
| JFK-ACC-SW03 | Vulnerable | no                             | <ul style="list-style-type: none"> <li>Config confirmed vulnerable: 2</li> <li>Config-independent vulnerability: 1</li> <li>Config detection not supported: 3</li> <li>Config not vulnerable: 7</li> </ul> | <ul style="list-style-type: none"> <li>Has known exploits: 3</li> <li>No known exploits: 10</li> </ul> | <ul style="list-style-type: none"> <li>High: 4</li> <li>Medium: 9</li> </ul> | 12+ months ago            | Cisco  |

## Devices Tab

The **Devices** tab is the default view when navigating the page. This tab applies a device-centric filter to the vulnerabilities list, focusing on summarizing CVE detection results for each impacted device.

Key metrics are displayed at the top while a filterable list of devices is displayed at the bottom.

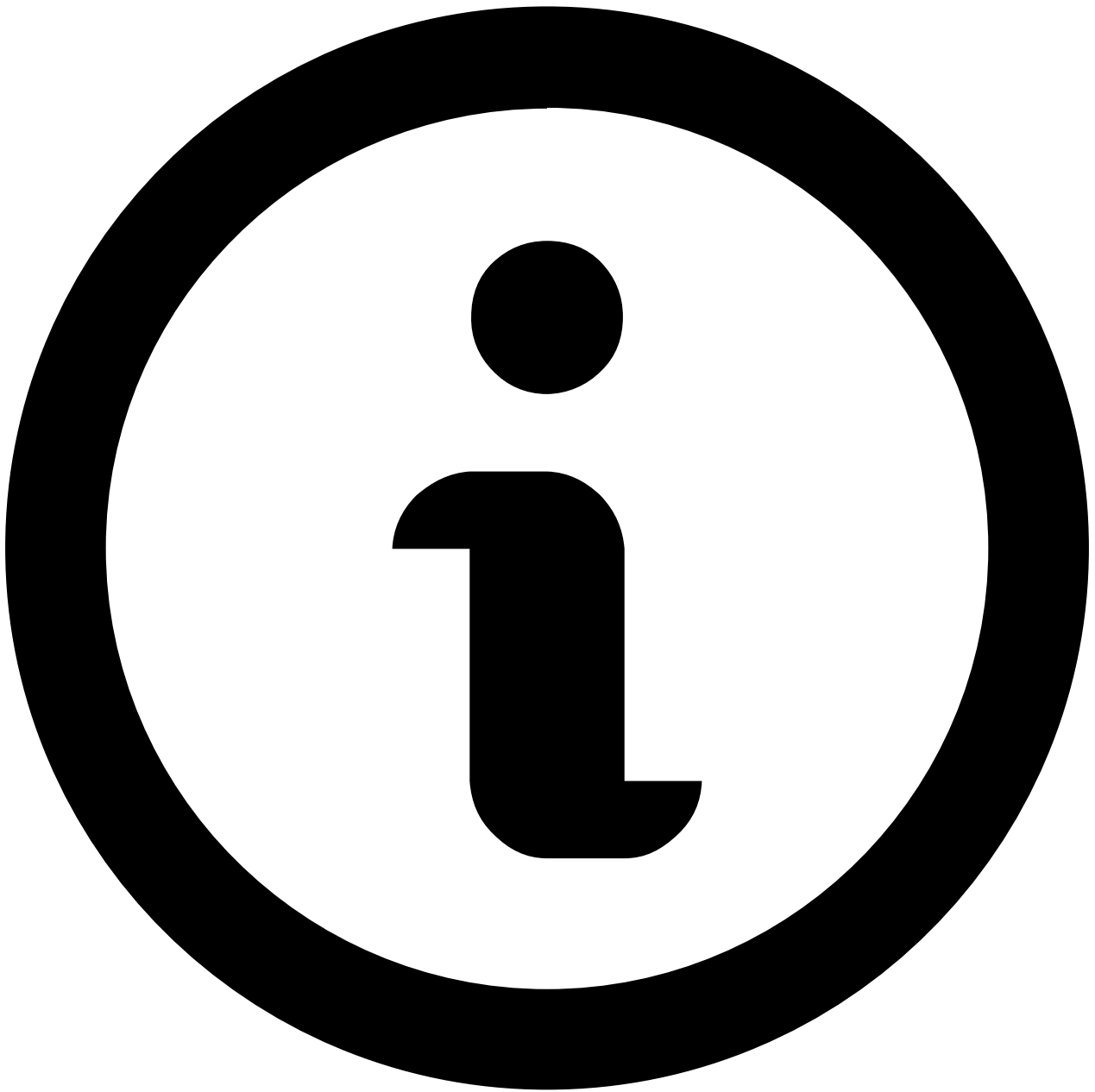


## Key Metrics

Key metrics displays numerical counts for device-related vulnerability data:

- **Total devices analyzed:** Total number of devices where the current OS version is affected by at least one CVE included in the analysis.
- **Devices affected by confirmed vulnerabilities:** Number of devices confirmed as vulnerable, with counts broken down by:
  - Devices with known exploits
  - Devices affected by CVEs with critical severity
  - Devices affected by CVEs published recently (last 30 days)
- **Devices receiving traffic from internet:** Number of devices receiving traffic from internet. Counts are broken down by:
  - Devices with known exploits
  - Devices with confirmed vulnerabilities

- Devices with unconfirmed vulnerabilities
- Devices affected by CVEs published recently (last 30 days)



note

The values above will be hidden if the Internet node is not configured.

Clicking any metric applies a corresponding filter to the displayed results.

## Checkbox CVE filters[a](#)[x](#)

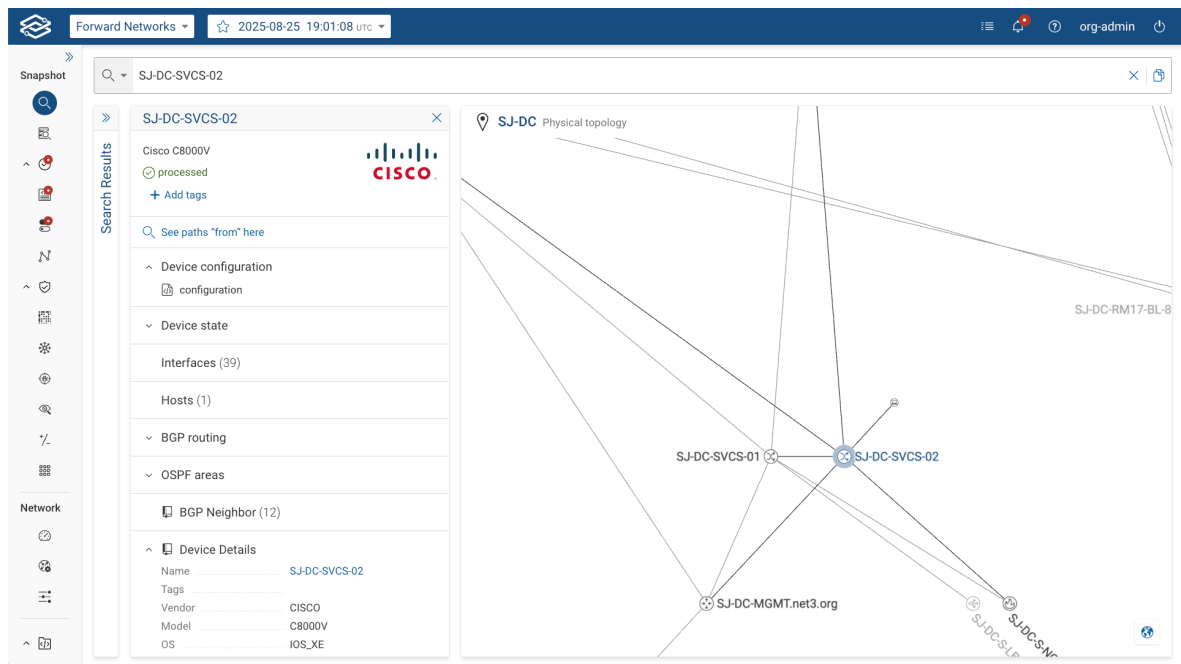
Checkbox filters select a subset of CVE findings before grouping them by device in the table below. Available checkbox filters include:

- CVEs with known exploits
- CVEs with critical severity
- CVEs published in the last 30 days

## Device List[a](#)[x](#)

Each row in the list of devices includes the following details:

- **Name:** The name of the device. Clicking the device name opens a new browser tab in **Search**, displaying additional details about the device for further investigation.



- **Status:** Displays a summary of the device status based on the currently applied filters: **Vulnerable**, **Potentially vulnerable**, **Config not vulnerable**.
- **Receives traffic from internet:** Indicates whether the device receives traffic from the Internet, which can impact its vulnerability risk.
- **CVE count by detection results:** Describes how the CVE was identified on the device. See [Detection Results](#) for details on result categories.
- **CVE count by known exploits:** Shows how many detected CVEs on the device have documented exploits. Filter options include **Has known exploits** and **No known exploits**.



- **CVE count by severity:** Lists the count of severities of the detected CVEs from **Critical** to **Low**.
- **CVE count by publish date:** Lists the counts of cves within each age bucket (past 30 days, 1 month to 12 months, older than 1 year).
- **Vendor:** The vendor of the device.
- **OS:** The operating system of the device.
- **OS Version:** The specific OS versions on the device.
- **Tags:** Tags associated with the impacted devices for easier identification and filtering.
- **Location:** Assigned location of the device.

Each column includes a table filter. Applying table filters will select all device rows where the filter value is present.

Clicking the **See details** icon opens a drawer with additional information about the selected device.

**Forward Networks** | 2025-05-15 13:50:54 | Overview

**LAX-MPLS-RT01**

Clear all filters | CVE filters | ☐ With known exploits ☐ Critical severity ☐ Published in the last 30 days

Location:  
LA (Los Angeles, California, United States)

[View device in search](#)

**Device details**

Vendor: CISCO  
Model: IOSv  
OS version: 15.5(3)M  
Management IPs: 172.31.1.21  
Internet addressable: No

**Status**

✖ Vulnerable

**Detection results**

✖ Config confirmed vulnerable: 3  
✖ Config independent vulnerability: 6  
⚠ Config detection not supported: 13  
ⓘ Config not vulnerable: 60

**CVE severities**

Critical: 1  
High: 60  
Medium: 21

82 detected CVEs | 0 hidden columns | [Download .csv](#)

| CVE ID                         | Detection result                   | Severity | CVE base score | Publish date | CWE                     | Description                                                               | Known exploits               | CVE age         |
|--------------------------------|------------------------------------|----------|----------------|--------------|-------------------------|---------------------------------------------------------------------------|------------------------------|-----------------|
| <a href="#">CVE-2018-0151</a>  | ✖ Config-independent vulnerability | Critical | 9.8 (v3.0)     | 2018-03-28   | <a href="#">CWE-119</a> | A vulnerability in the quality of service (QoS) subsystem of Cisco IOS... | <a href="#">CISA sourced</a> | 12+ months ago  |
| <a href="#">CVE-2025-20352</a> | ⓘ Config not vulnerable            | High     | 7.7 (v3.0)     | 2025-09-24   | <a href="#">CWE-121</a> | A vulnerability in the Simple Network Management Protocol...              | <a href="#">CISA sourced</a> | 1-12 months ago |
| <a href="#">CVE-2025-20253</a> | ⚠ Config detection not supported   | High     | 8.6 (v3.0)     | 2025-08-14   | <a href="#">CWE-835</a> | Multiple vulnerabilities in the Internet Key Exchange Version 2...        | none                         | 1-12 months ago |
| <a href="#">CVE-2025-20239</a> | ⚠ Config detection not supported   | High     | 8.6 (v3.0)     | 2025-08-14   | <a href="#">CWE-401</a> | Multiple vulnerabilities in the Internet Key Exchange Version 2...        | none                         | 1-12 months ago |
| <a href="#">CVE-2025-20182</a> | ⚠ Config detection not supported   | High     | 8.6 (v3.0)     | 2025-05-07   | <a href="#">CWE-787</a> | A vulnerability in the Internet Key Exchange...                           | none                         | 1-12 months ago |

[Close](#)

The drawer displays:

- **Device details:** Displays device-specific information.
- **Status:** Displays the number of detection results for configs confirmed vulnerable and configs confirmed not vulnerable.
- **CVE severities:** Displays the count of CVE severities detected on the device, broken into categories: Critical, High, Medium.

The drawer also provides a table of CVEs affecting the device, with the following details for each CVE:

- **CVE ID:** The unique identifier for each CVE. Clicking this opens detailed CVE information.
- **Detection result:** Describes how the CVE was identified on the device. See [Detection Results](#) for details on result categories.
- **Severity:** The severity rating of the CVE.
- **CVE base score:** The CVSS base score of the CVE.
- **Published date:** The date when the CVE was officially published.
- **CWE:** Associated CWE identifiers, hyperlinked to the [MITRE CWE database](#).
- **Description:** A brief description of the CVE and its impact.
- **Known Exploits:** Shows if the CVE has a published exploit. Entries display the source (CISA or vendor). CISA entries link directly to the [CISA Known Exploited Vulnerabilities Catalog](#).
- **CVE age:** Displays the age of the CVE.

## CVEs Tab

Selecting the **CVEs** tab applies a CVE-centric filter to the vulnerabilities list.

Key metrics are displayed at the top while a filterable list of CVEs is displayed at the bottom.

Demo Network

2024-04-09 18:27:26 UTC

noreply@noreply.com

Snapshot

Search

Inventory

Verify

Security

Posture

Blast Radius

Vulnerability

Exposure

Diffs

Objects

Debug

Network

Dashboard

Sources

Collection

Onboarding

Platform

NQE

Collectors

Settings

Forward Docs

Vulnerability

CVE database version in use: 2026-01-14 15:16

The possible impact of CVEs is analyzed based on the device OS version, configuration, reachability from the internet, and other factors. Learn more in the [Forward Documentation](#)

Update CVE database

Devices

CVEs

Key Metrics

376

Total CVEs detected

191

confirmed vulnerabilities

with known exploits

6

impacting devices receiving traffic from internet

89

published recently (last 30 days)

0

13

unconfirmed vulnerabilities

with known exploits

1

impacting devices receiving traffic from internet

8

published recently (last 30 days)

0

Clear all filters

Device filters

☐ Receives traffic from internet

Locations:

Tags:

376 detected CVEs

0 hidden columns

Download .csv

| CVE ID         | Severity (high...) | Known exploits | Description                                                                                                                                                   | Device count by detection result                        | Vendor  | OS               | OS version | Confi  |
|----------------|--------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------|------------------|------------|--------|
| Filter         | Filter             | Filter         | Filter                                                                                                                                                        | Filter                                                  | Filter  | Filter           | Filter     | Filter |
| CVE-2025-60010 | Medium             | none           | A password aging vulnerability in the RADIUS client of Juniper Networks Junos OS and Junos OS Evolved allows an authenticated, network-based attacker to ...  | <div>Config detection not supported</div> <div>16</div> | Juniper | Juniper Junos OS | 14.1R5.4   | Yes, c |
| CVE-2025-59980 | Medium             | none           | An Authentication Bypass by Primary Weakness in the FTP server of Juniper Networks Junos OS allows an unauthenticated, network-based attacker ...             | <div>Config not vulnerable</div> <div>16</div>          | Juniper | Juniper Junos OS | 14.1R5.4   | Yes    |
| CVE-2025-59962 | Medium             | none           | An Access of Uninitialized Pointer vulnerability in the routing protocol daemon (rpd) of Juniper Networks Junos OS and Junos OS Evolved with BGP sharding ... | <div>Config not vulnerable</div> <div>16</div>          | Juniper | Juniper Junos OS | 14.1R5.4   | Yes    |

An Improper Neutralization of Delimiters vulnerability in the IP of Juniper Networks

## Key Metrics

Key metrics display numerical counts for CVE-related data:

- **Total CVEs detected:** Overall number of CVEs affecting at least one device in the network.
- **confirmed vulnerabilities:** Number of CVEs confirmed to impact devices, with counts broken down by:
  - with known exploits
  - affecting devices receiving traffic from internet
  - recently published (last 30 days)
- **unconfirmed vulnerabilities:** Number of CVEs that may impact devices, but where [Enhanced Analysis](#) beyond OS version matching was not performed, with counts broken down by the same categories as the confirmed vulnerabilities.

Clicking any metric applies a corresponding filter to the displayed results.

## Checkbox Device filters

Checkbox filters select a subset of device vulnerability findings before grouping them by CVE in the table below. Available checkbox filters include:

- (devices that) Receive traffic from the internet
- (device) Locations
- (device) Tags

## CVE List

The CVEs list includes:

- **CVE ID:** The unique identifier for each CVE. Clicking this opens detailed CVE information.
- **Severity (highest):** The severity rating of the CVE, indicating how critical the vulnerability is.
- **Known exploits:** Shows if a CVE has a published exploit. Entries display the source (CISA or vendor). CISA entries link directly to the [CISA Known Exploited Vulnerabilities Catalog](#).
- **Description:** A brief description of the CVE and its impact.

- **Device count by detection result:** Describes how the CVE affects devices in the network with corresponding **Detection Results** counts. See [Detection Results](#) for details on result categories.
- **Vendor:** The vendors of the devices affected by the CVE.
- **OS:** The operating systems affected by the CVE.
- **OS version:** The operating system versions affected by the CVE.
- **Config dependent:** True if the CVE depends on a specific device configuration, unknown if enhanced analysis was not performed on the CVE.
- **Devices impacted:** The number of devices with each of the OSes affected by the CVE.
- **CVE Base score:** CVE CVSS based score for that OS.
- **Published date:** The date when the CVE was officially published.
- **Tags on impacted devices:** Tags associated with the impacted devices for easier identification and filtering.
- **Locations of impacted devices:** The assigned location of the impacted devices.
- **CWE:** Common Weakness Enumeration number and link for this CVE

Each column includes a table filter. Applying table filters will select all CVE rows where the filter value is present.

Clicking the **View details** icon opens a drawer with additional information about the selected CVE. The drawer will display a separate tab for each affected Operating system. In each tab the drawer displays:

- **Severity:** The severity rating of the CVE.
- **CWE classification:** Associated CWE identifiers, hyperlinked to the [MITRE CWE database](#).
- **CVE base score:** The CVSS base score of the CVE.
- **Published date:** The date when the CVE was officially published.
- **Description:** A brief description of the CVE and its impact.
- **Config dependent:** Indicates whether enhanced analysis is supported and CVE is configuration-based.
- **Vendor / OS:** The vendor and affected operating systems.
- **Vendor advisory:** A link to the vendor's published advisory, if available.

The screenshot displays the Cisco Security Center interface for CVE-2025-20352. The left sidebar shows navigation options like Snapshot, Search, Inventory, Verify, NQE, Predefined, Intent, Advanced, Security, Posture, Blast Radius, Vulnerability, Exposure, Diffs, Objects, and Debug. The main panel shows the CVE details, including its severity (High), CWE classification (CWE-121), base score (7.7), and published date (2025-09-24). Below the details is a table of impacted devices.

| Device                    | Status         | Detection result            | Receives traffic from internet | Location   | Tags | Model          | OS version |
|---------------------------|----------------|-----------------------------|--------------------------------|------------|------|----------------|------------|
| phx-bldg8-cEdge           | Not vulnerable | Config not vulnerable       | No                             | Unassigned |      | C8000V         | 17.12.02   |
| poe02.dc.forwardnetwor... | Not vulnerable | Config not vulnerable       | No                             | Unassigned |      | WS-C3850-48U-S | 16.09.05   |
| SJ-BLDG1-C9800-WLC10      | Vulnerable     | Config confirmed vulnerable | No                             | Unassigned |      | C9800-CL-K9    | 17.13.01   |
| SJ-BLDG1-SP-EDGE-01       | Not vulnerable | Config not vulnerable       | No                             | Unassigned |      | C8000V         | 17.09.01a  |
| SJ-BLDG1-SP-EDGE-02       | Not vulnerable | Config not vulnerable       | No                             | Unassigned |      | C8000V         | 17.09.01a  |
| SJ-DC-SVCS-01             | Not vulnerable | Config not vulnerable       | No                             | Unassigned |      | C8000V         | 17.09.01a  |

The drawer also provides a table of impacted devices, with the following details for each device:

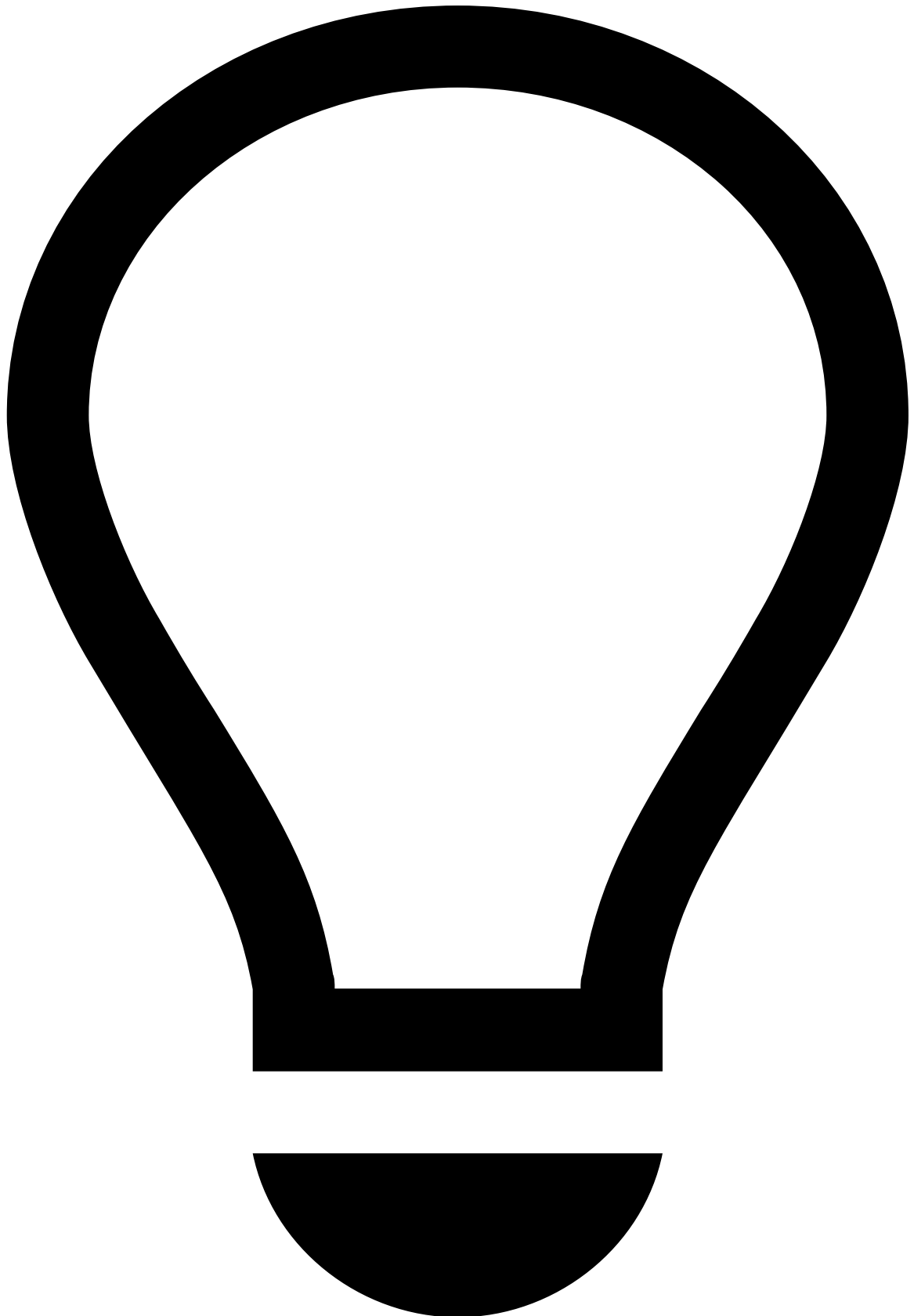
- **Device:** The name of the impacted device. Clicking this will provide more detailed information about the device.
- **Status:** classifies the finding as Vulnerable, Possibly Vulnerable, or Not Vulnerable.
- **Detection Result:** Describes how the CVE was detected on the device. Configuration icon on the right of the **Detection Result** opens a drawer highlighting the lines in the device configuration that make this vulnerability exploitable. For CVEs that affect multiple versions of OSes, there are multiple lines (one per OS) for the following columns: **Vendor**, **OS**, **OS version**, **Devices impacted**, **Tags**, and **Locations**. The information in each lines corresponds to the affected devices configured with that Operating System.
- **Receives traffic from Internet:** Indicates whether the device receives traffic from the Internet.
- **Locations:** The assigned location of the impacted devices.
- **Tags:** Device tags.
- **Model:** The model of the impacted device.
- **OS version:** The operating system version running on the impacted device.
- **Management IPs:** The management IP addresses of the impacted device.

## Data Filtering

Clicking on any of the **Key Metrics** applies the corresponding filter to the Device or CVE list table.

The same behavior applies when filtering by **Locations** or **Tags** in the CVEs view.

You can add additional filters directly in the list tables using the filters available for each column.



#### tip

When applying both **Key Metrics** and column-based filters, it is recommended to apply the **Key Metrics** filter first. This is because **Key Metrics** filtering reduces the underlying data set, while column-based filtering only refines how that data is viewed.

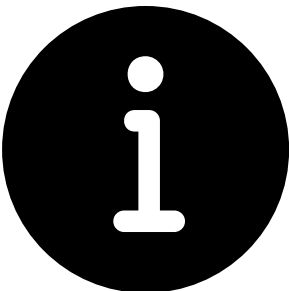
You can remove all filters by clicking **Clear all filters** in the filter section.

**Detection Results**[ââ](#)

The detection result indicates how a device's vulnerability status was determined. Results fall into one of five categories:

| Result                                                                                                                                                                         | Description                                                                                      | Recommended actions                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <div><br/><b>Vulnerable: Configuration-independent vulnerability (Platform)</b></div>         | The operating system version itself is affected, regardless of configuration.                    | Apply a patch or up/downgrade to a non-vulnerable OS version.                                                                     |
| <div><br/><b>Vulnerable: Config confirmed vulnerable (Platform and configuration)</b></div> | The OS version is affected, and the device configuration confirms that the CVE is exploitable.   | Apply a patch, up/downgrade to a non-vulnerable OS version, or modify the section of the device configuration affecting this CVE. |
|                                                                                                                                                                                | The OS version is affected, but configuration-based detection is not yet supported for this CVE. | If possible, apply a patch; analyze the device configuration manually to see if a                                                 |



| Result                                                                                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                        | Recommended actions                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|  <p><b>Potentially Vulnerable: Config detection not supported (Platform)</b></p>              |                                                                                                                                                                                                                                                                                                                                                    | configuration work-around is possible.                                                                                     |
|  <p><b>Potentially Vulnerable: Unable to confirm configuration (Platform)</b></p>            | The OS version is affected, but Forward could not confirm whether the configuration makes the device vulnerable.                                                                                                                                                                                                                                   | Recheck this device after the next snapshot. If the error persists, report a bug to Forward and apply a patch if possible. |
|  <p><b>Not Vulnerable: Config confirmed not vulnerable (Platform and configuration)</b></p> | A CVE is reported against this OS version, but the configuration required for the device to be vulnerable is not present. In this case, the OS may contain the vulnerability, but the device's configuration prevents exploitation. Devices that are not vulnerable at the platform level do not appear in the Vulnerability Analysis application. | Plan to upgrade the device OS during a regular maintenance window.                                                         |

In the UI, results are color-coded by severity:

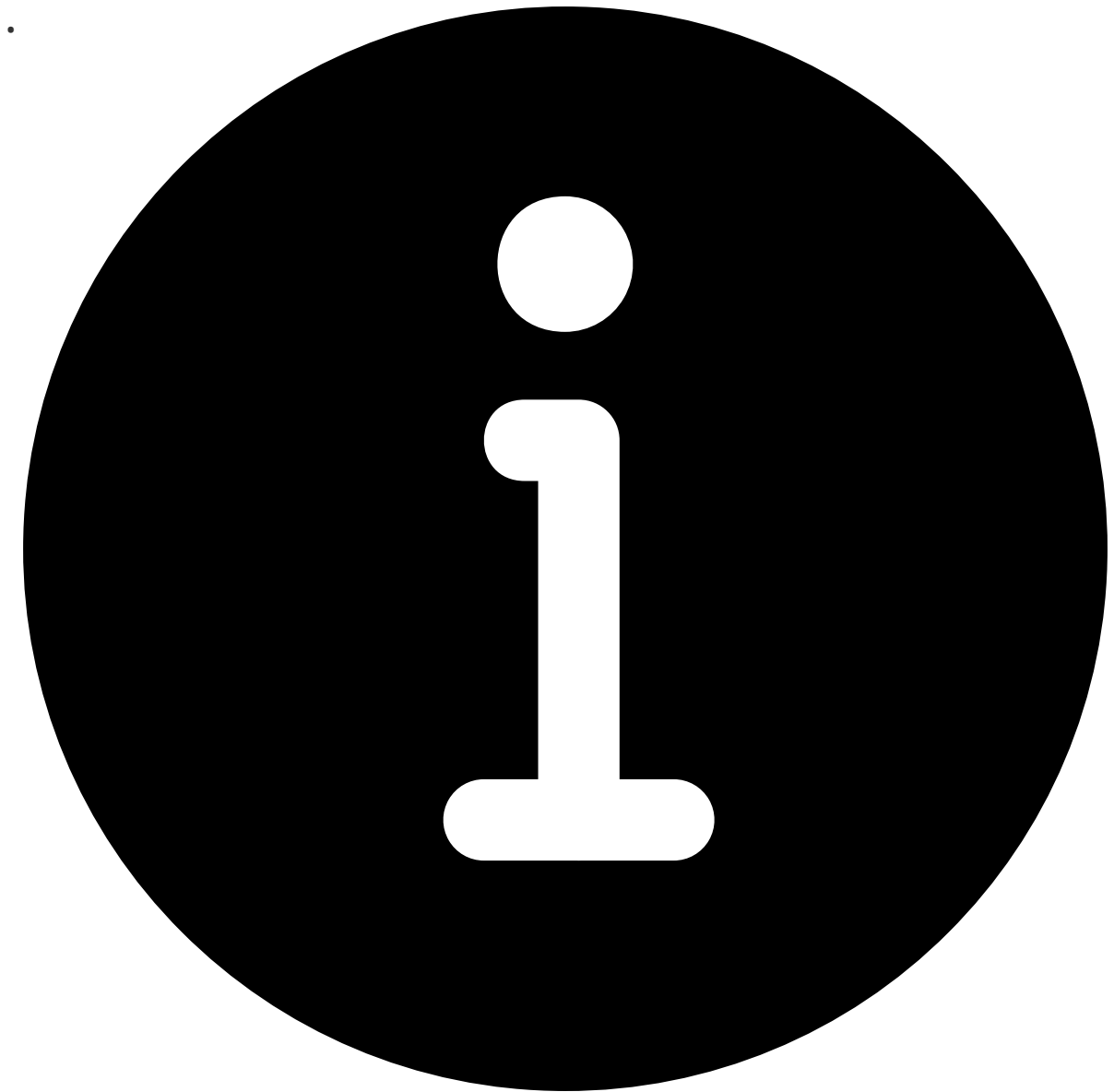
- 



**Vulnerable** (red)



**Potentially Vulnerable** (orange)



**Not Vulnerable** (black)

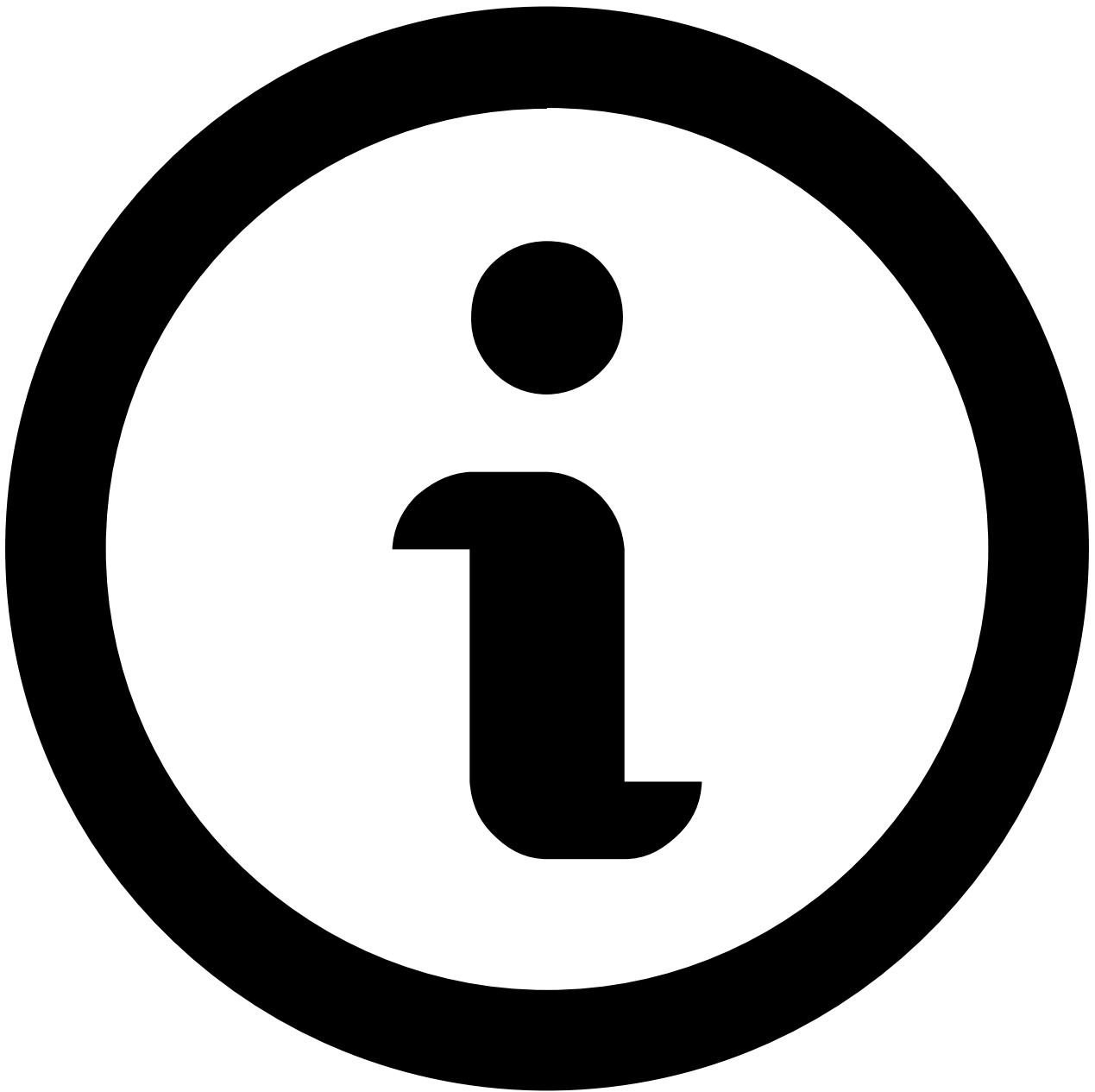
## **CVE Search**[🔍](#)

To search for a CVE, enter the CVE ID in the vulnerabilities table. The search can result in:

- At least one device is affected by the vulnerability.
- CVE-ID does not affect any device supported by Forward Enterprise
- The CVE ID is not present in the Forward CVE database (it may be outdated, irrelevant to supported vendors, or outside the date range included).

## CVE Database Update [a](#)

On Forward SaaS, the CVE database is updated daily. For self-hosted deployments, the CVE database is updated with each Forward Enterprise release, and it can also be updated using the methods described below.



note

The list of CVEs considered by Forward Networks may exclude hardware vulnerabilities or CVEs that were not created or updated after January 1, 2014.

## Scheduled

Starting with release 26.2, you can schedule automatic hourly updates of the CVE database. This ensures that newly published vulnerabilities that pose a high risk to the enterprise are included in the vulnerability analysis.

To enable scheduled updates, navigate to **Settings > System > Software Central**, and enable the **Enable hourly updates** toggle. If a newer version is detected, the system automatically downloads and installs it.

Scheduled updates require that the Forward Cluster has network access to [Forward Software Central](#). Use **Test Connection** to verify connectivity and authentication settings. If necessary, configure a proxy, including options to validate certificates and set proxy authentication.

The **Update now** option checks for the latest version of the vulnerability database and installs it immediately, without waiting for the next scheduled hourly update.

You can see the latest CVE database version in use at the bottom of the page.

**Forward Software Central**

 Although Forward Network software is hosted locally, several features are enriched by periodically connecting to Forward Networks at <https://fwd.app>. Check the tooltips below for each feature's purpose and the data exchanged during the connection session.

### Connection settings

☐ Enable proxy [?](#)

URL

https://fwd-proxy-

☐

Verify certificate

Tip: Disable if proxy uses a self-signed certificate.

☐

### Enable authentication

## Test connection

connectivity verified

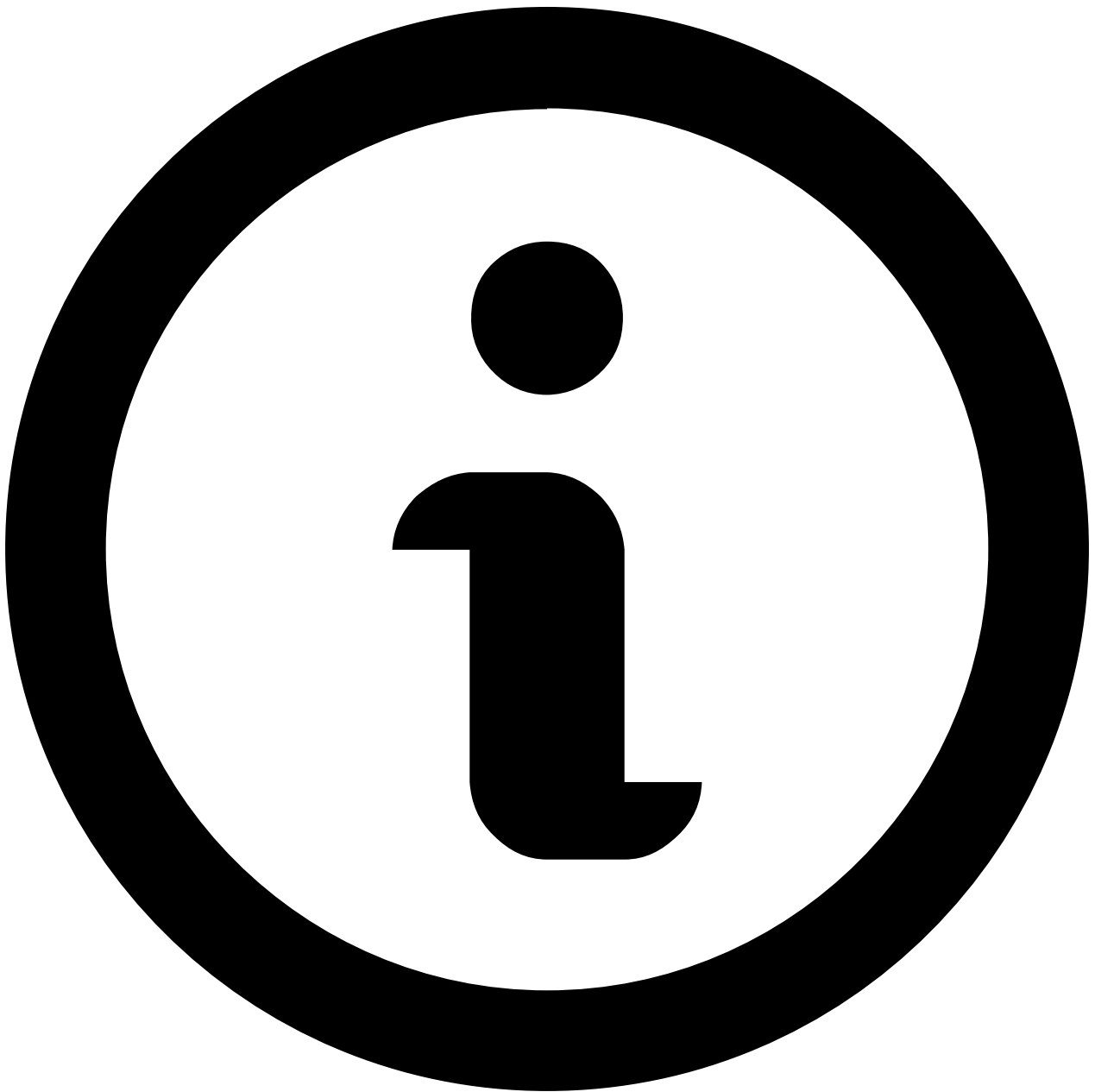
CVE database

☒ Enable hourly updates [?](#)

[Update now](#)

②

CVE database version in use: 2026-02-17 08:50:23 GMT-05:00



note

**Test Connection** verifies Deployment ID and the active license ID as well to ensure the request is valid.



## Manual

Although scheduled updates are the recommended approach, you can manually update the CVE database:

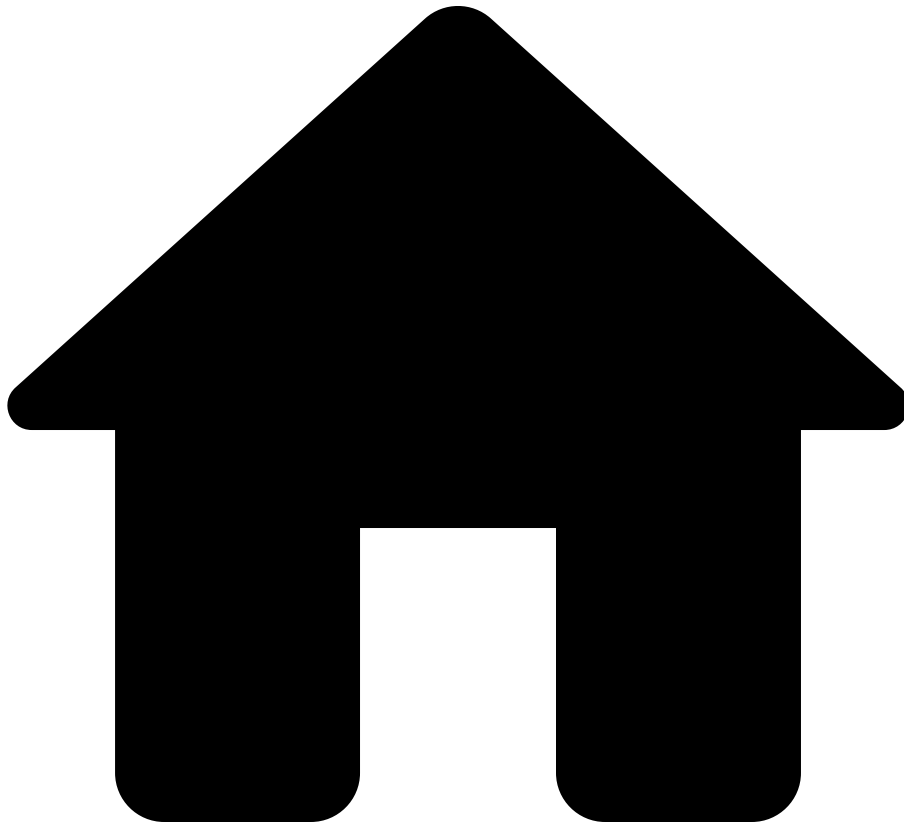
1. Download the latest CVE database from the [CVE Update](#) page.
2. Navigate to **Security > Vulnerability** from the left navigation menu.
3. Click **Update CVE database** (top right corner) to upload the CVE database.  
Optionally, enter the **Checksum** to verify the file's integrity.
4. Click **Update**.

## REST APIs

1. **Download the CVE Database:** Any user with *Org Admin* privileges can download the latest CVE database from *Forward Software Central* using the Forward REST APIs via `GET /api/cve-index`. It returns the CVE database as a gzipped binary (.bin.gz) file.
2. **Update the CVE Database:** A user with *Org Admin* privileges can update the CVE database using the Forward REST APIs via `PUT /api/cve-index`.

For detailed instructions, see the [REST API docs](#).

•



- [NQE](#)
- Decorators

On this page

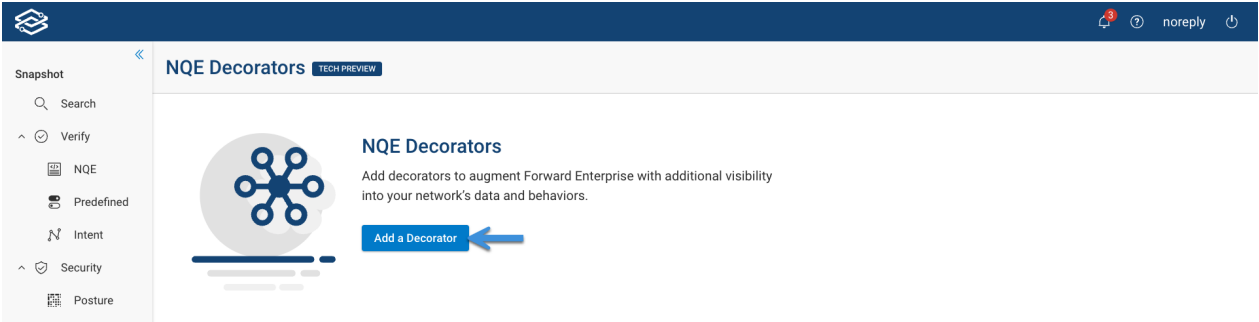
## Decorators

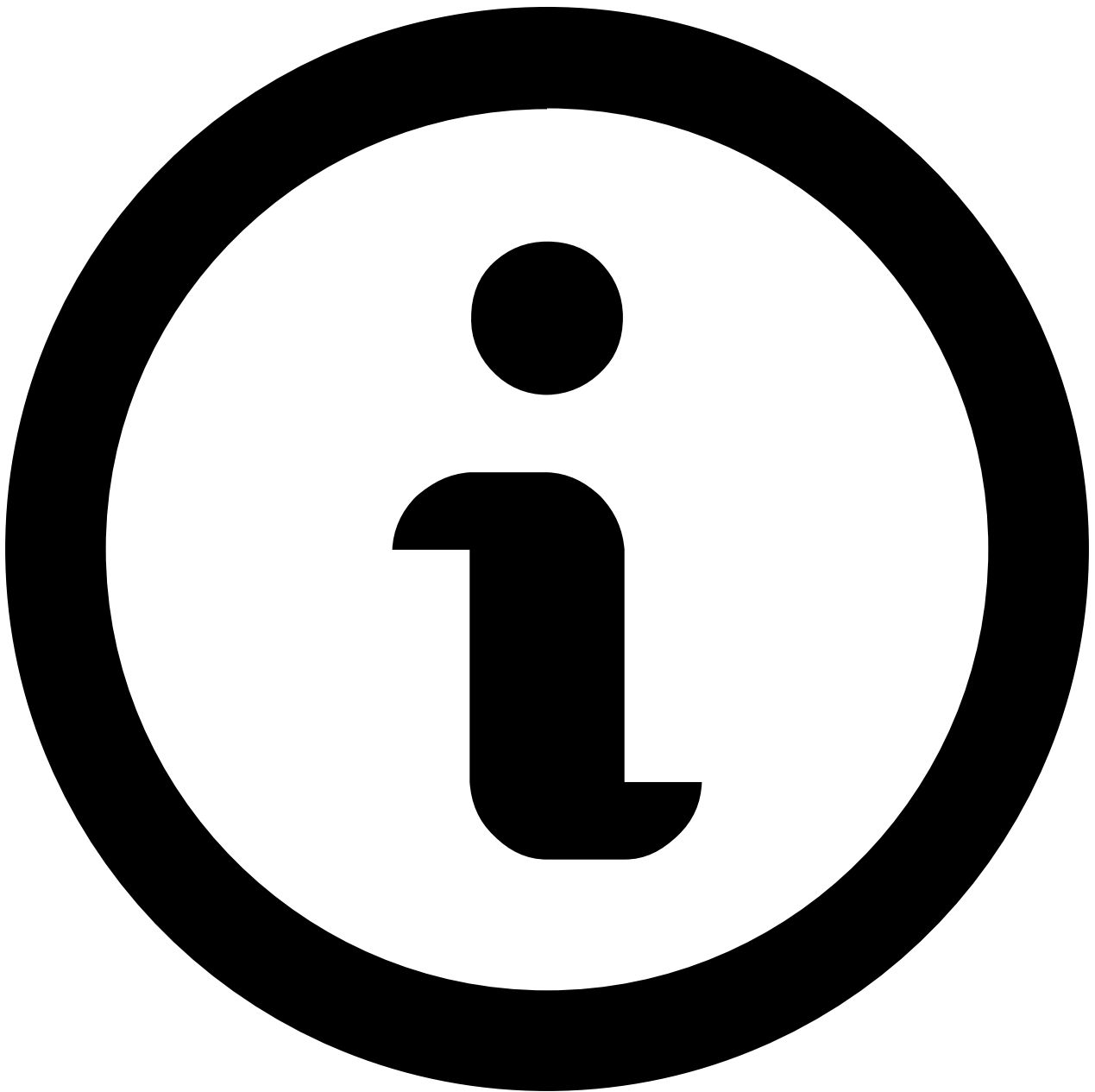
**NQE Decorators** enable the use of [NQE Queries](#) to enhance Forward applications by incorporating additional contextual data.

Forward supports NQE Decorators for **Devices, Hosts, Interfaces, and Paths**.

# Add a Decorator

To add a Decorator, navigate to **NQE > Decorators** and click **Add a Decorator**

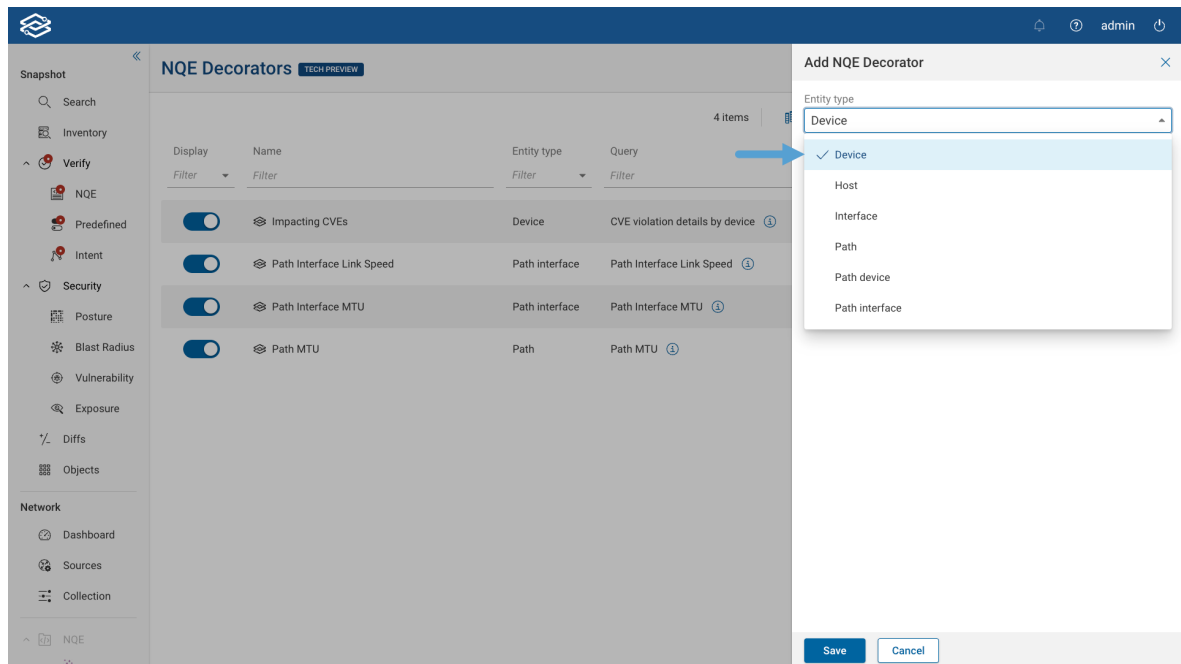




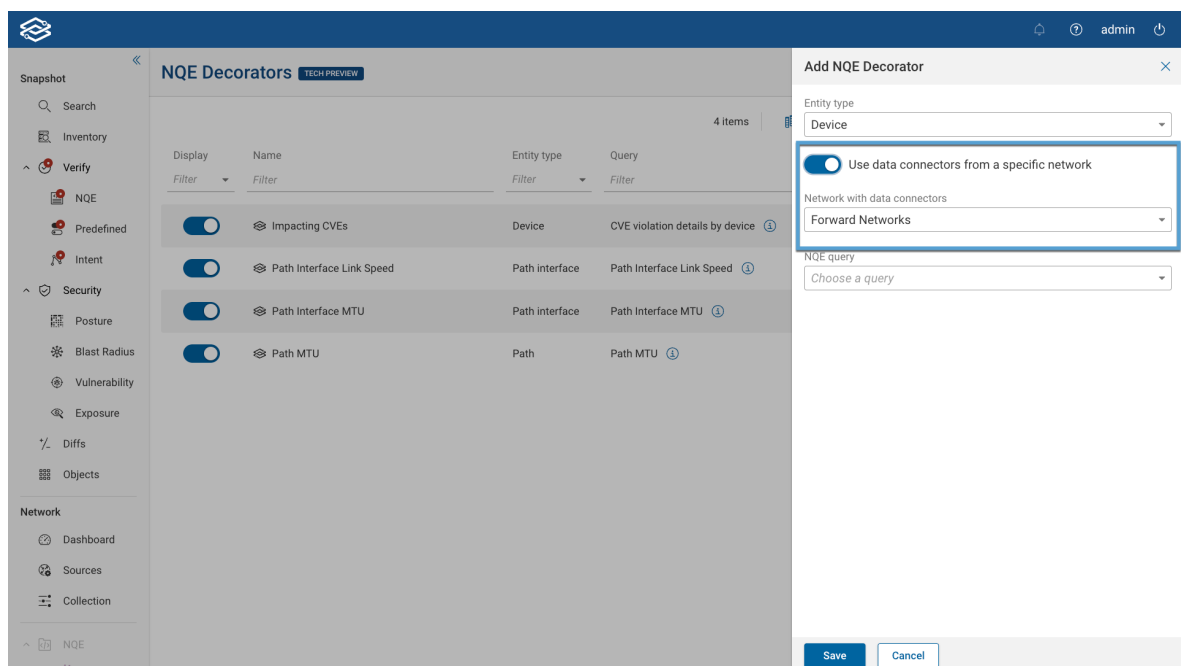
note

The drawer for adding an NQE Decorator is very similar for devices, hosts, interfaces, and paths. The following describes the flow for devices.

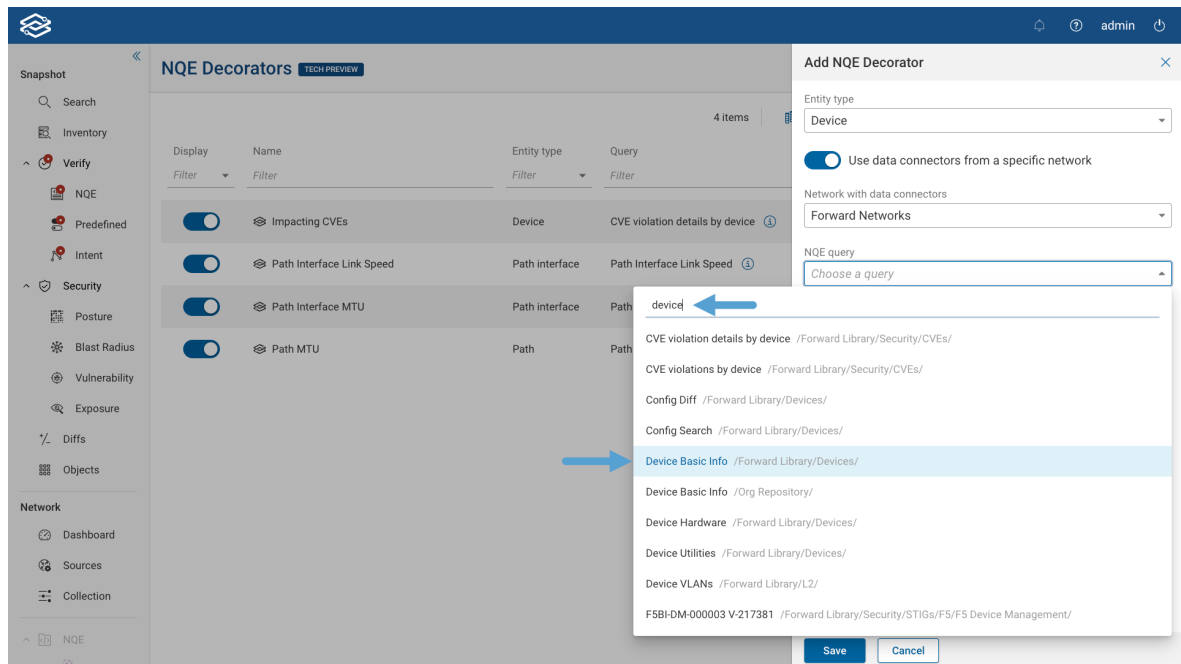
- Select **Device** from the **Entity type** drop-down menu



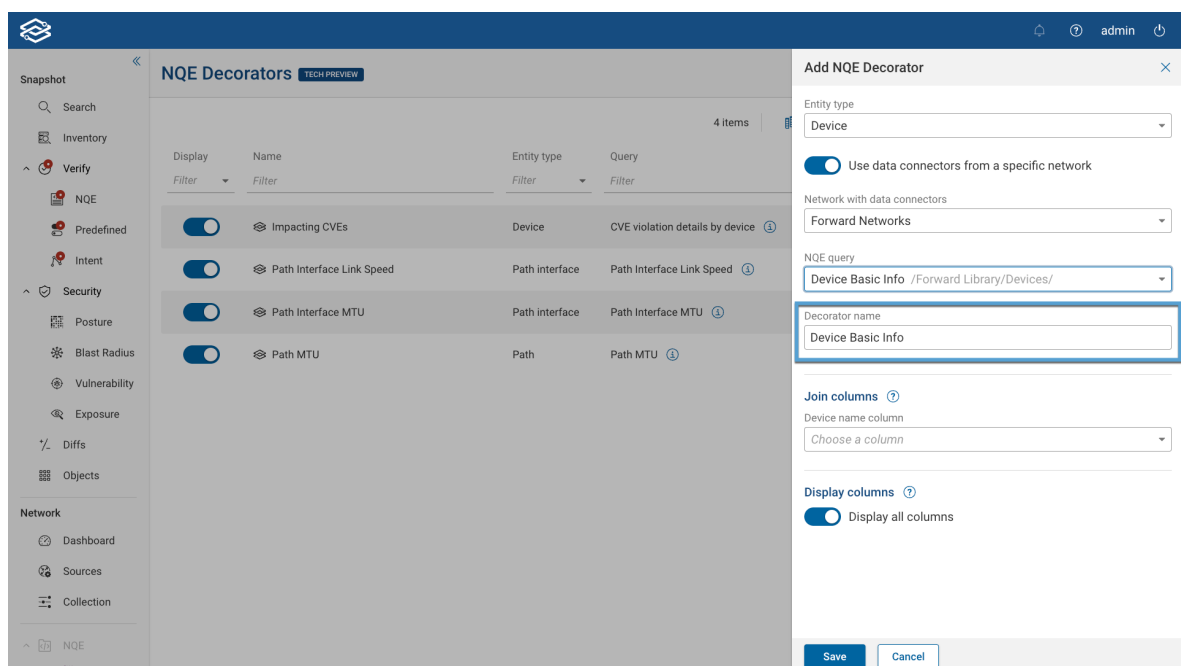
- **Use data connectors from a specific network:** If the NQE Decorator relies on an NQE query that retrieves data from a [data connector](#), enable this option and choose the Network with the data connectors from the provided drop-down menu.



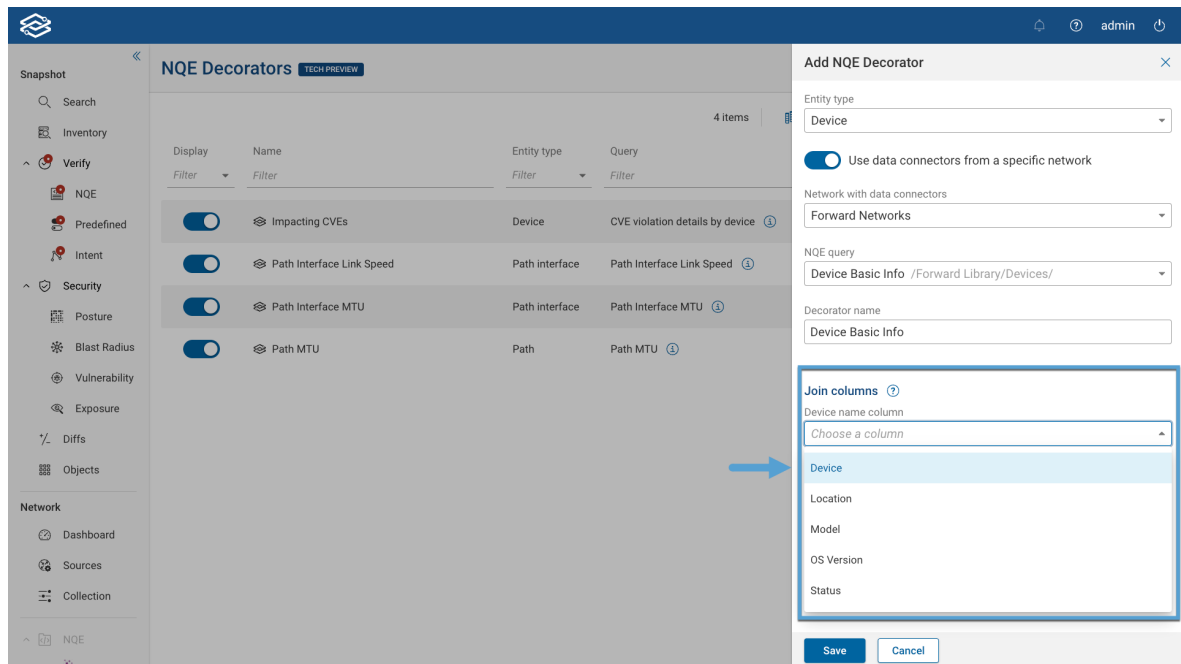
- **NQE query:** choose an NQE query from the drop-down menu. Start typing the NQE query name to quickly find it among the full list of queries.



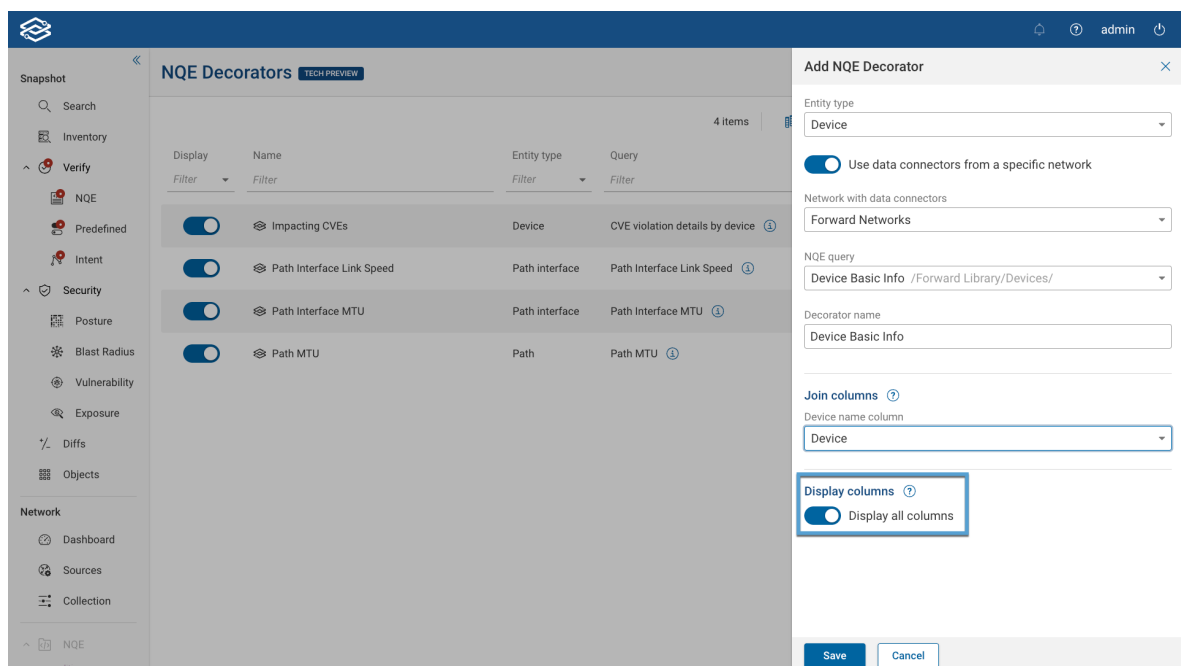
- The **Decorator name** field is pre-filled with the NQE query name, but it can be customized.



- **Join Columns:** The columns from the NQE query results to use to filter the results before joining/combining them with an individual entity. For example, you must choose the column that contains device names to decorate devices. For each device being decorated, the chosen column will be used to filter the results to those that match the device.

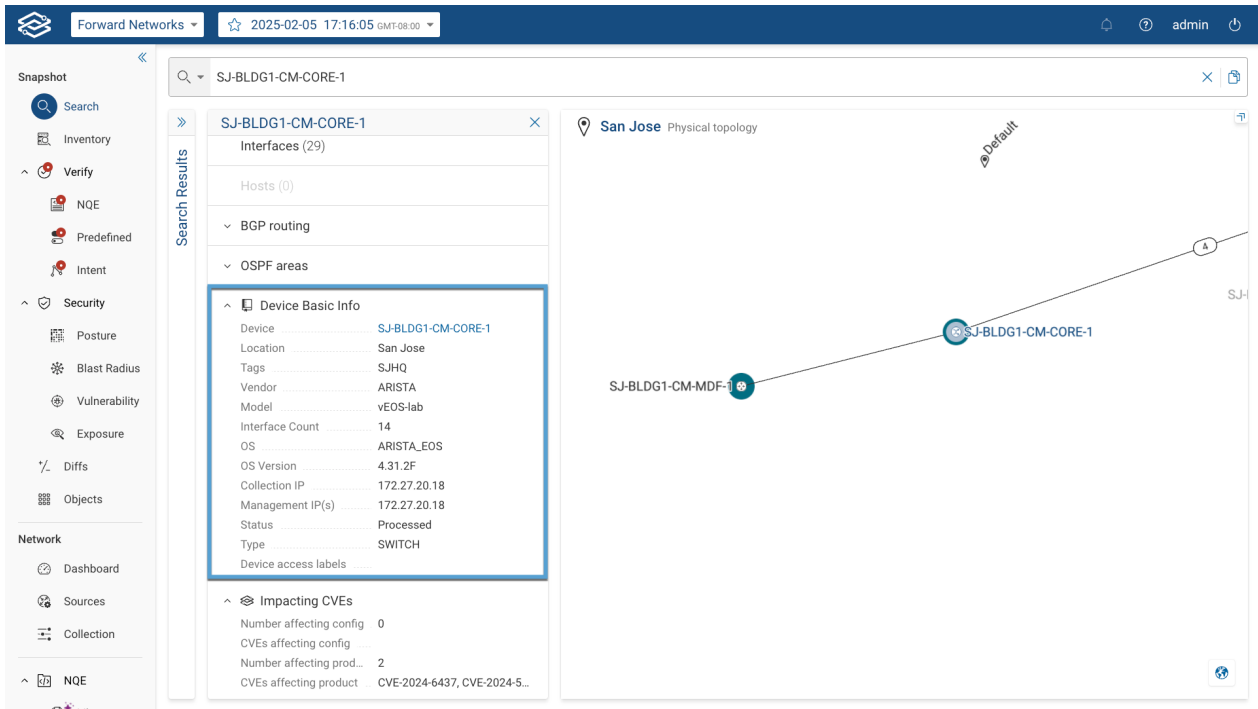


- **Display all columns:** Keep this selected to display all columns from the query's results when displaying this Decorator's data. Turn this off to choose specific columns to display instead.



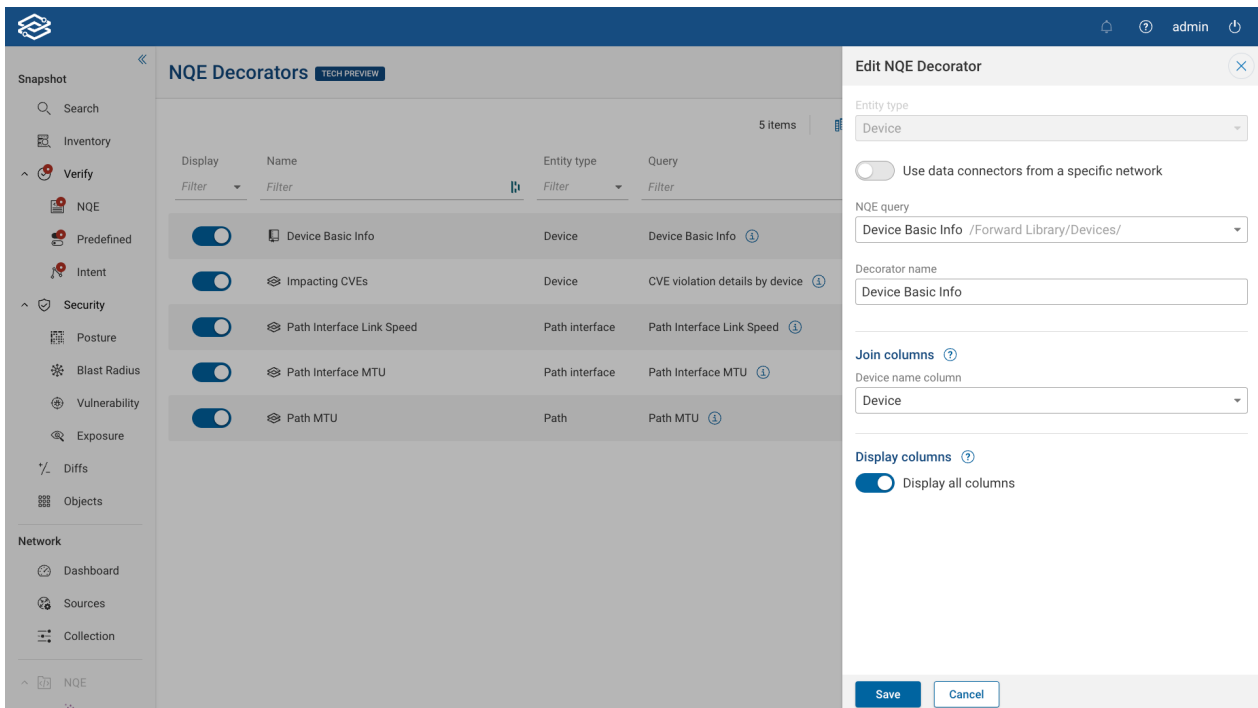
Then, click on **Save**.

The following screenshot shows a device card with an example NQE Decorator.



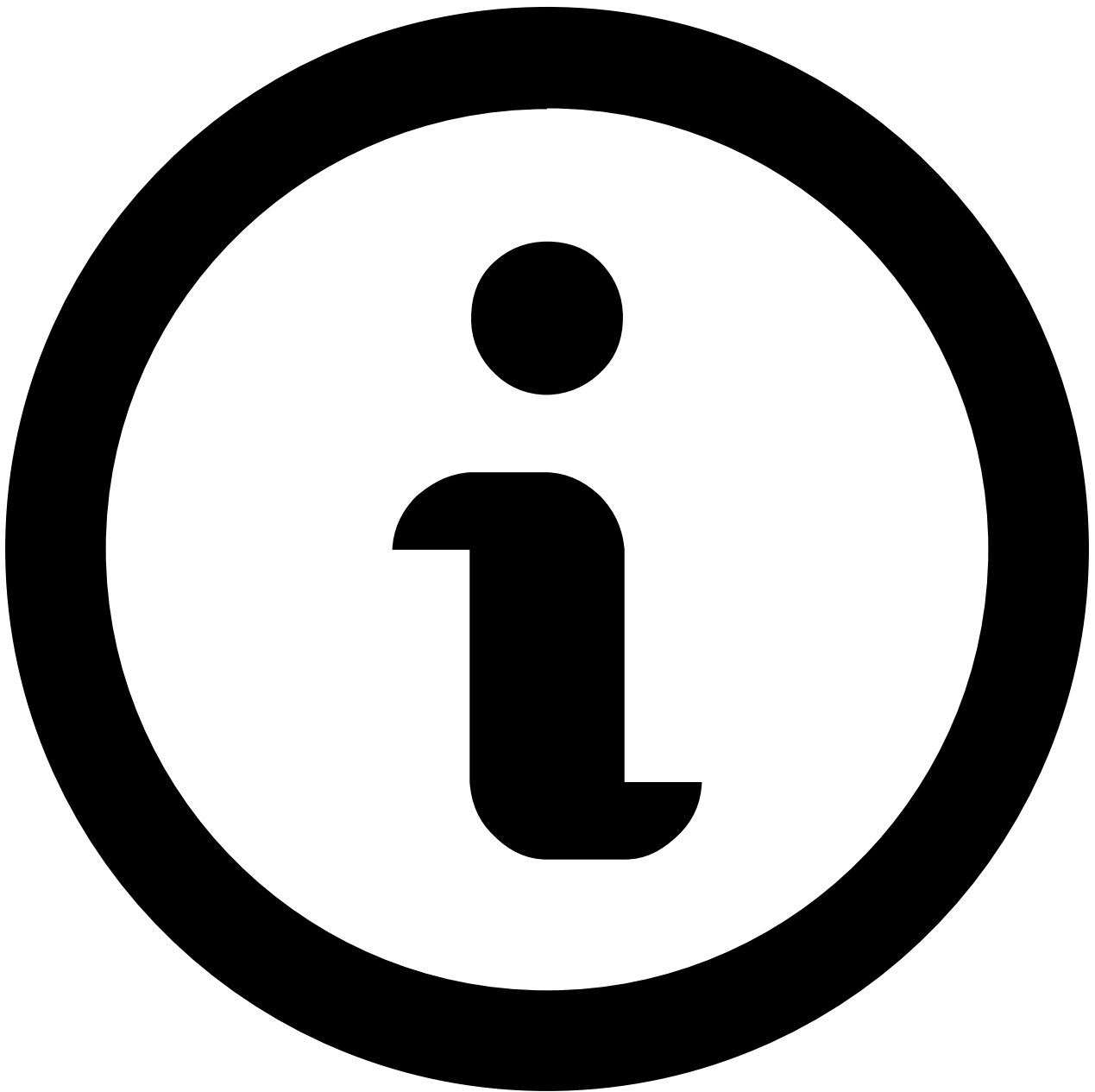
## Display/Edit/Delete a Decorator

Decorators can be turned off/on by using the **Display** toggle in the table.



To edit an NQE Decorator, click on the edit icon on the right side of the Decorator's row in the table, change any Decorator detail, and click on **Save**.

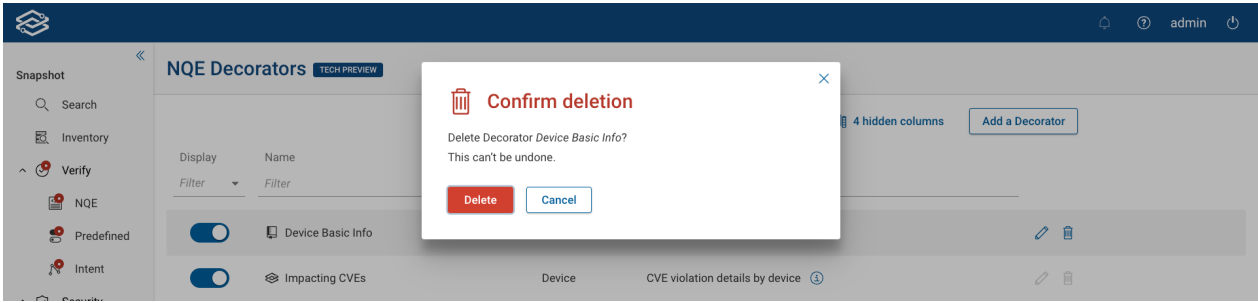




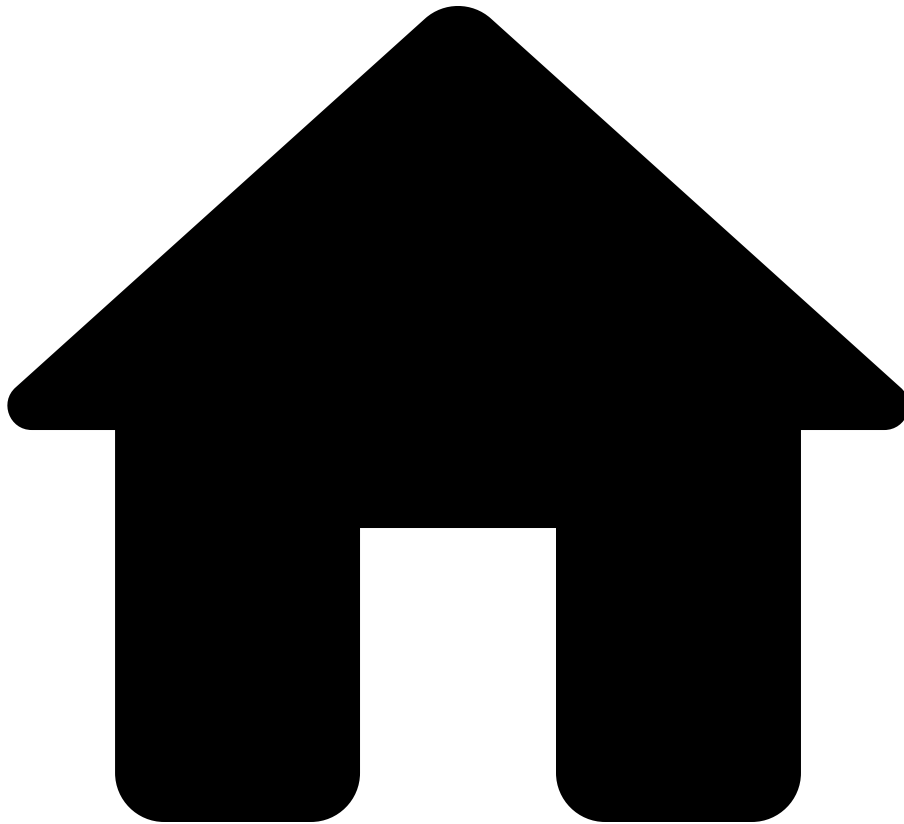
note

Once a Decorator is created, it is not possible to change the **Entity type** associated with it.

To delete an NQE Decorator, click on the bin icon on the right side of the Decorator's row in the table, and confirm its deletion.



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- [Verify](#)
- NQE Verifications

On this page

## NQE Verifications

### Overview[ââ](#)

NQE stands for **Network Query Engine**. NQE enables you to write **queries** for information about your network. The queries return a collection of records that are displayed as rows in a table.

An **NQE Verification** is formulated as a query for violations. Every NQE Verification follows a common pattern: query the data model to identify violations of the property that you want to verify. If the query finds no violations, then the verification passes. If the query finds violations, the verification is marked as failed, and the identified violations form the failure diagnosis. You can create, store, run, diagnose and publish your **NQE queries** entirely in the Forward Enterprise [NQE Library](#) application. The NQE queries need to be published in the NQE Library application via the **Commit** process in order to be used as NQE Verifications.

## Getting started

Follow instructions in the editor to get started writing your first check. Some key points to notice:

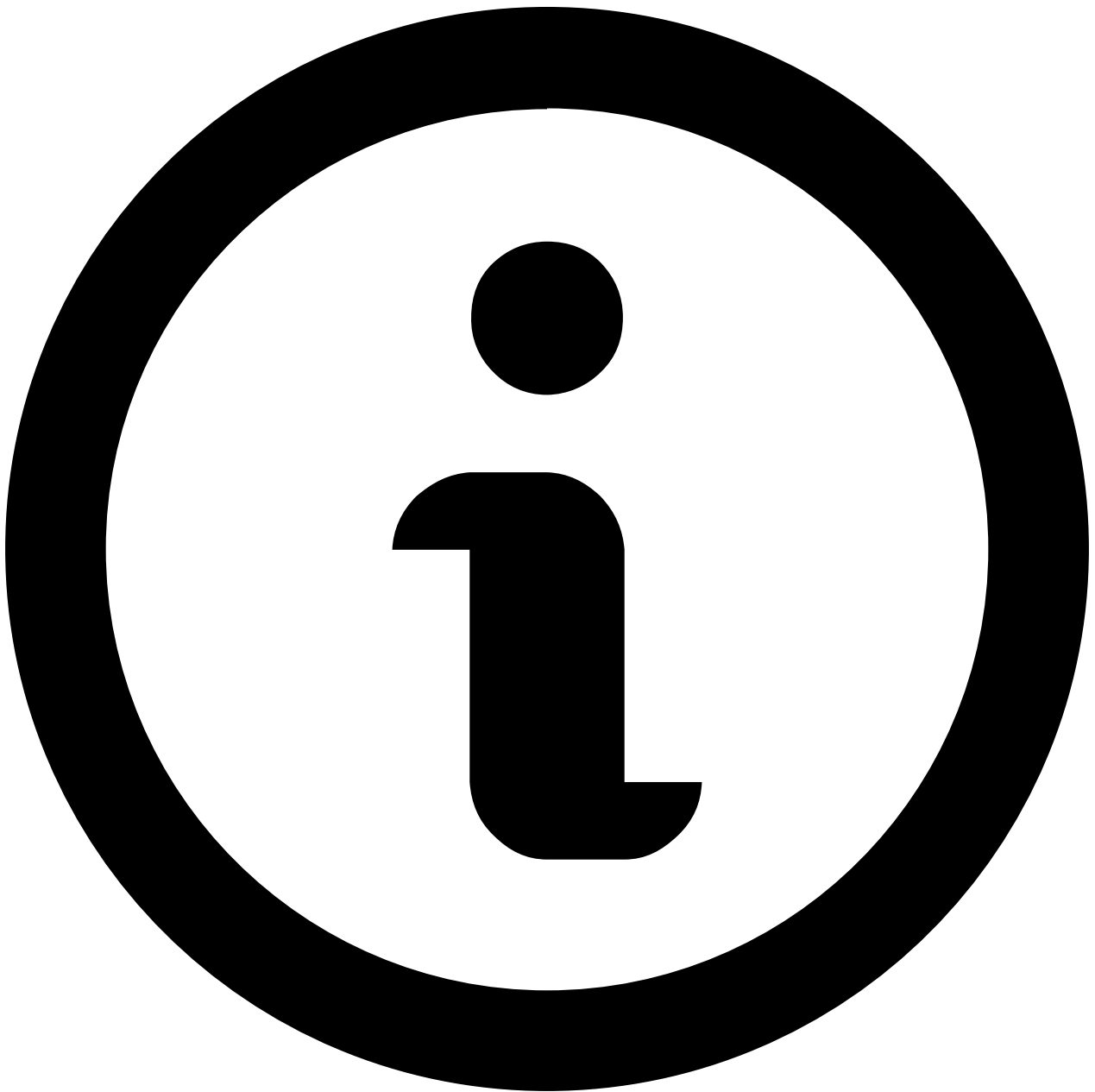
- Results show up below in the *Results* tab.
- If there are problems with your edits, the editor will underline problematic parts. Hover over the underlined part to see a detailed description and suggestion for how you might fix the problem.
- The [Overview section](#) describes the key parts of an NQE Check.

## NQE Checks with `violation` label

An **NQE Check** is formulated as an NQE query. Each row may include the label `violation` containing a value of type `Bool` indicating whether the row is a violation or a passing verification. The rows of an NQE Check should include all attributes of the objects of interest that are relevant to understanding why the row is labeled as a violation or a passing verification. This allows readers of the check results to quickly understand what the NQE check is verifying and why an item may be failing.

If any rows returned by an NQE query are labeled as violations, then the check is marked failed and the attributes in each violation row forms the diagnosis for the failure indicated by the row. Otherwise, the check passes, and the rows can be used to inspect what network items were positively verified.

The numbers of passing and failing rows are charted on the Dashboard as the number of passing verifications and the number of violations, respectively.



note

The letters in the `violation` label must be all lowercase. If some letter is uppercase, then a warning will be given.

### **NQE Checks with no `violation` label**[↗](#)

If an NQE Check does not include a `violation` label to indicate whether each row is violation or passing verification, it is assumed that every row is a violation. In addition, for such an NQE Check, if the query produces no rows, then the number of passing

verifications is 1 and the number of violations is 0. Otherwise, the number of passing verifications is 0 and the number of violations is 1.

Note that this type of check is deprecated and allowed for backwards-compatibility. Support for this type of check will be removed in a future release.

## Anatomy of a Check

NQE Checks use a syntax similar to [LINQ](#), which is itself similar to SQL.

For example, the following check finds every interface on Cisco NXOS devices. It marks each interface that is admin UP, but not operationally UP as a violation. All other interfaces are considered passes.

```
foreach device in network.devices
foreach interface in device.interfaces
where device.platform.os == OS.NXOS
select {
  deviceName: device.name,
  interfaceName: interface.name,
  adminStatus: interface.adminStatus,
  operStatus: interface.operStatus,
  violation: interface.adminStatus == AdminStatus.UP &&
             interface.operStatus != OperStatus.UP
}
```

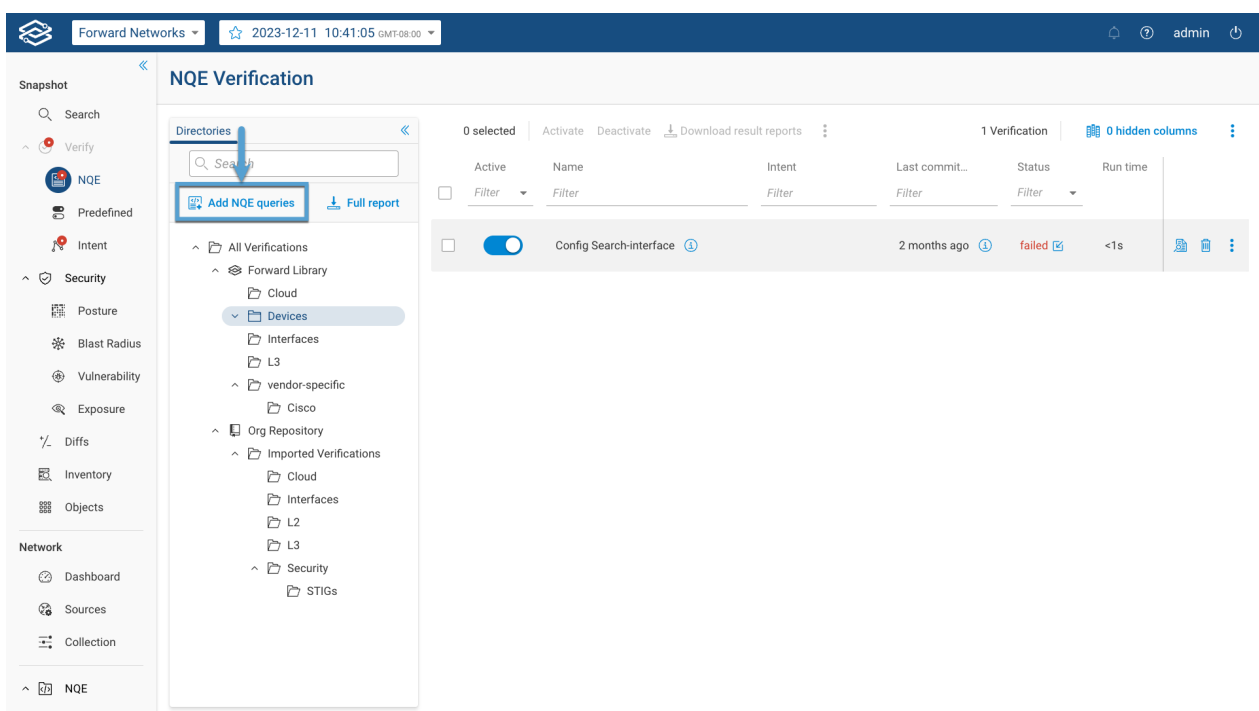
The basic idea is that `foreach` clauses iterate over the data, the `where` clauses restrict (filter) the data to the condition you care about, and the `select` clause determines what information is included in the diagnosis. Expanding on this:

- **foreach** : In the above check, the first two `foreach` clauses iterate over the devices and their interfaces. `foreach` only applies to collections of elements. There can be any number of `foreach` clauses.
- **Scoping**: `foreach` introduces a new variable that is available in the subsequent lines of the check. In the above example, `device` and `interface` are introduced in the first two lines and then used in the last few lines of the check.
- The only variable in scope initially is `network`. This variable provides access to all attributes of the network.

- **where** : The **where** clause limits the data to just that matching the condition you care about. Think of it as a way to filter data. The **where** clause supports any boolean-valued condition, including expressions constructed from familiar boolean and relational operators, such as **&&** (and), **||** (or), **!** (not), **==** , **!=** , **<** , **>** , etc. You can have any number of **where** clauses. They can appear intermixed with **foreach** clauses, so long as they only use variables in scope at that line.
- **select** : The final **select** clause constructs a record (or object) for each combination of **device** and **interface** values that satisfy the **where** condition. These records determine what information is included in the diagnosis. The field names are used as column names in the results table. There can be only one **select** clause.
- **violation** field: Each row produced in the **select** clause should have a field called **violation** with a boolean-valued condition. This condition defines whether a row is considered a violation or a passing verification.
- A record is a comma-separated list of name-value pairs enclosed in curly braces: `{name1: value1, ...}`.
- Case-sensitivity: The language is case-sensitive.
- Whitespace: The language is not sensitive to the number of spaces or line breaks.

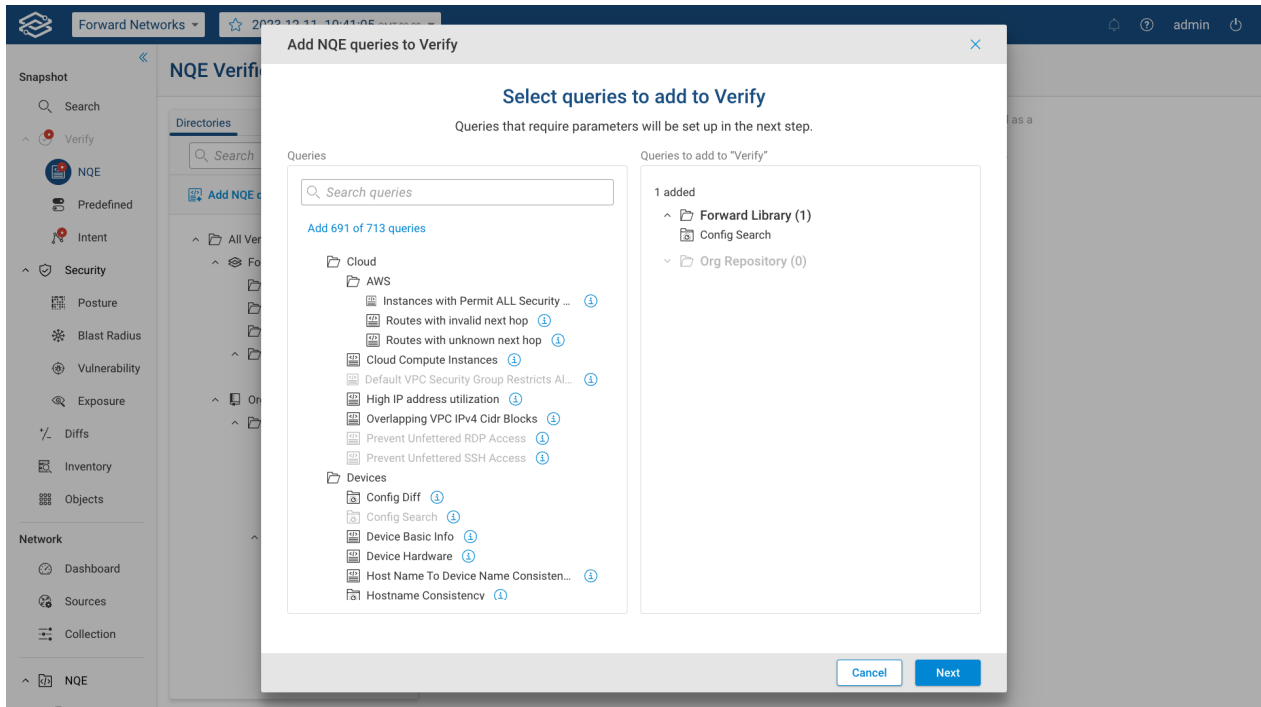
## Adding NQE Queries to NQE Verifications

To add NQE queries to NQE Verifications, initiate the process by selecting the **Add NQE Queries** button within the NQE Directory Panel.



## Selecting NQE Queries

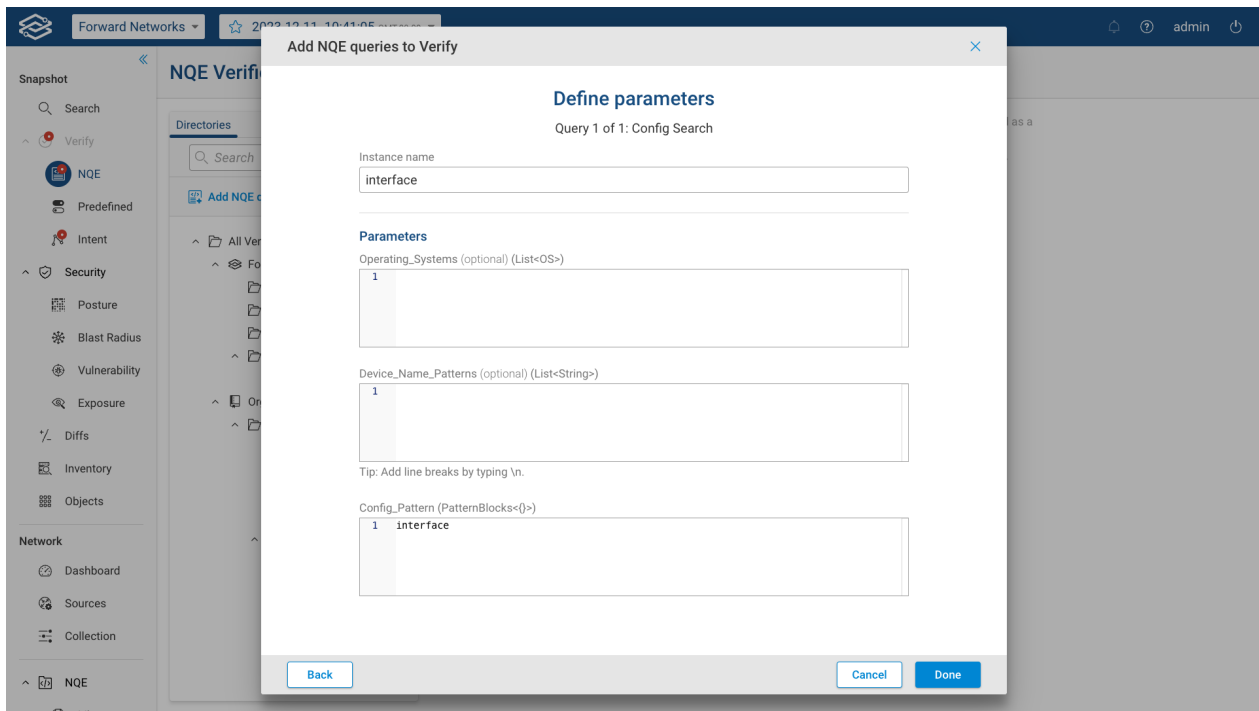
Choose one or more queries from the available directories to add as NQE Verifications. The selected queries will be listed in the adjacent panel.



## Defining Query Parameters

For NQE queries requiring additional parameters to be configured, define them before adding. For each query, provide a Name and enter the preferred parameter values. This process repeats until all NQE query parameters are assigned.

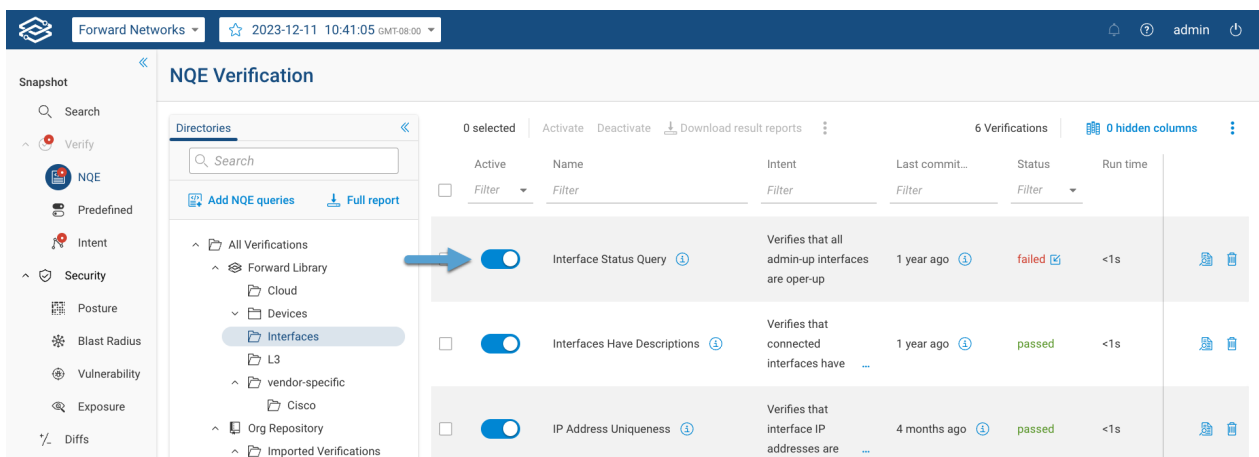




Visit the [Parameterized Queries](#) section of the [NQE Library](#) page for additional information on parameterized queries.

## Enabling/Disabling NQE Verifications

By default, newly added NQE queries start in an enabled (**active**) state. Clicking on the **Active** toggle of an NQE Verification will change its state to disabled (**inactive**).



## Deleting NQE Verifications

To delete an NQE Verification, select the **Delete** icon, located at the end of each row, to delete it.

# NQE Verification Results

If an NQE Verification fails, the violations and query are available by clicking on the **Failed** status link, or by selecting the **View Details** icon. The **Results** tab displays the report, while the Code tab displays the NQE query used to generate the report.

Forward Networks

2023-12-11 10:41:05 GMT-08:00

admin

Snapshot

Search

Verify

NQE

Predefined

Intent

Security

Posture

Blast Radius

Vulnerability

Exposure

Diff

Inventory

Objects

Network

Dashboard

Sources

Collection

NQE

Interface Status Query

ResultsCode

112 of 112 results

Download XLSX report

Download CSV report

| violation | deviceName                                  | interfaceName                               | adminStatus | operStatus |
|-----------|---------------------------------------------|---------------------------------------------|-------------|------------|
| Filter    | Filter                                      | Filter                                      | Filter      | Filter     |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | vgw-2f2ccd46/dxvif-fg7zr93m.1               | UP          | UP         |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | bdr-102-2.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | vgw-34f41f5d/dxvif-ffq5gw3.2                | UP          | UP         |
| false     | bdr-102-2.mgmt.dc2-iad.github.net.wan.1 ... | bdr-102-2.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | vgw-2f2ccd46/dxvif-fh7abxm0.1               | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | bdr-103-1.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | vgw-34f41f5d/dxvif-ffqcs70.2                | UP          | UP         |
| false     | bdr-103-1.mgmt.dc2-iad.github.net.wan.1 ... | bdr-103-1.mgmt.dc2-iad.github.net/ae3.1 ... | UP          | UP         |
| true      | MPLS                                        | LAX-MPLS-RT01/gi0/0                         | UP          | DOWN       |

Close

# Previewing Results with Updated Data Files

Some NQE-based verifications depend on external data files. When a data file is updated, the verification's details view includes a selector that allows you to run the check using:

- **Snapshot** the authoritative snapshot-based result
- **Snapshot with Latest Data Files** a preview that overlays the most recently uploaded data files onto the snapshot

This allows you to validate updates to data files immediately, without waiting for a new snapshot collection. The main verification status remains based on the snapshot.

Software Compliance

Results

Code

Data files

From Snapshot

Latest

32 of 32 filtered results

Download report

Device

OS

OS Version

Compliance Status

violation

Target Versions

Filter

Filter

Filter

Filter

Pass (17), ...

Filter

PHX-BLDG7-DIST-1.net3.org

ARISTA\_EOS

4.31.2F

✓ TARGET

Pass

4.31.2F

PHX-BLDG7-DIST-2.net3.org

ARISTA\_EOS

4.31.2F

✓ TARGET

Pass

4.31.2F

SJ-BLDG1-C9800-WLC10

IOS\_XE

17.13.01

⚠ ACCEPTABLE

Pass

17.12

SJ-BLDG1-CM-CORE-1

ARISTA\_EOS

4.31.2F

✓ TARGET

Pass

4.31.2F

SJ-BLDG1-SP-EDGE-01

IOS\_XE

17.09.01a

⚠ ACCEPTABLE

Pass

17.12

SJ-BLDG1-SP-EDGE-02

IOS\_XE

17.09.01a

⚠ ACCEPTABLE

Pass

17.12

Close

## STIG Compliance

Forward includes a built-in **STIG Findings** verification in the Forward Library, allowing you to assess compliance against supported DISA STIG rules.

In addition to the main STIG Findings check, the Library also provides **severity-specific STIG Findings queries** (for example, queries scoped to Critical, High, or Medium severity findings).

These checks can be add from Forward Library directly to the Verify page and behave like other NQE-powered verifications.

NQE Verification

Directories

Add NQE queries

Full report

All Verifications

Forward Library

Cloud

Interfaces

L3

Security

STIGs

vendor-specific

Org Repository

0 selected

Activate

Deactivate

Remove

Recompute

4 Verifications

1 hidden column

| Active                   | Name                    | Intent                                                                   | Verification status | Priority |
|--------------------------|-------------------------|--------------------------------------------------------------------------|---------------------|----------|
| Filter                   | Filter                  | Filter                                                                   | Filter              | Filter   |
| <input type="checkbox"/> | STIG Findings           | Verify compliance with all Forward-supported STIG rules.                 | failed              | medi...  |
| <input type="checkbox"/> | STIG Findings - Cat I   | Verify compliance with all high severity Forward-supported STIG rules.   | failed              | high     |
| <input type="checkbox"/> | STIG Findings - Cat II  | Verify compliance with all medium severity Forward-supported STIG rules. | failed              | medi...  |
| <input type="checkbox"/> | STIG Findings - Cat III | Verify compliance with all low severity Forward-supported STIG rules.    | failed              | low      |

## STIG Security Policy (User-Defined Parameters)

Some STIG rules require operator-supplied parameters to determine whether certain conditions apply to a device or interface.

These parameters are provided through the **STIG Security Policy** file ( `fn-stig-policy.csv` ), imported as a data file.

You can download the policy template from:

**NQE Data Files Import data file Download Template**

Import data file

Import data file

Upload your data file to begin the import process. Make sure that:

- .json or .csv extensions are in the file name
- STIGs policy file must follow a specific format. Download and fill it out the provided template before uploading it.

[Download template](#)

Note: Only one STIG policy file can exist in the system at any given time. Uploading a new policy file will replace an older version.

Drag and drop file here

The import process will start immediately after you select the file.

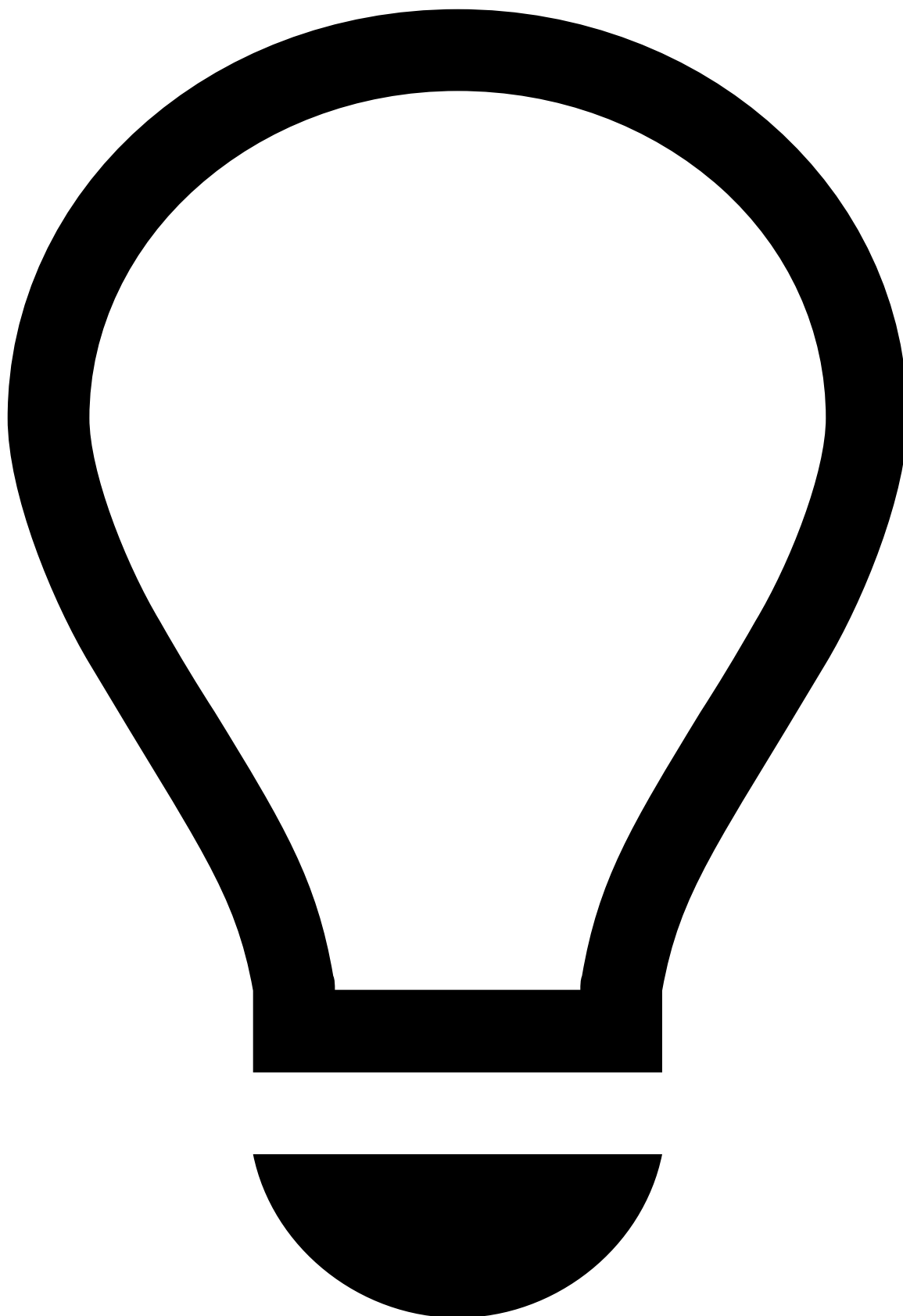
or

Select file

Cancel

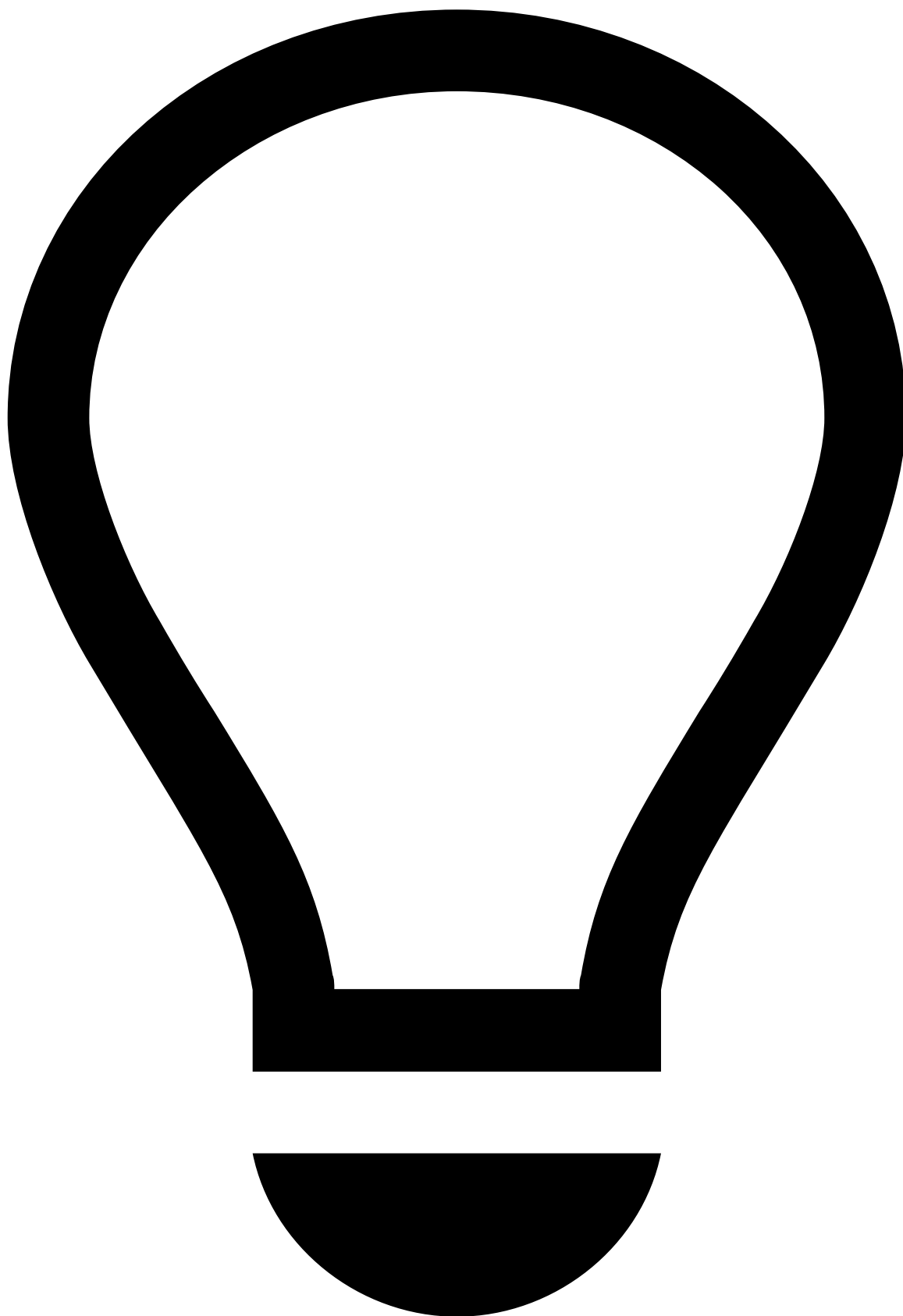
Next

After editing the policy values, upload the file using **NQE &#x2013; Data Files**. Once uploaded and enabled for the network, the STIG Findings verification automatically incorporates these parameters.



tip

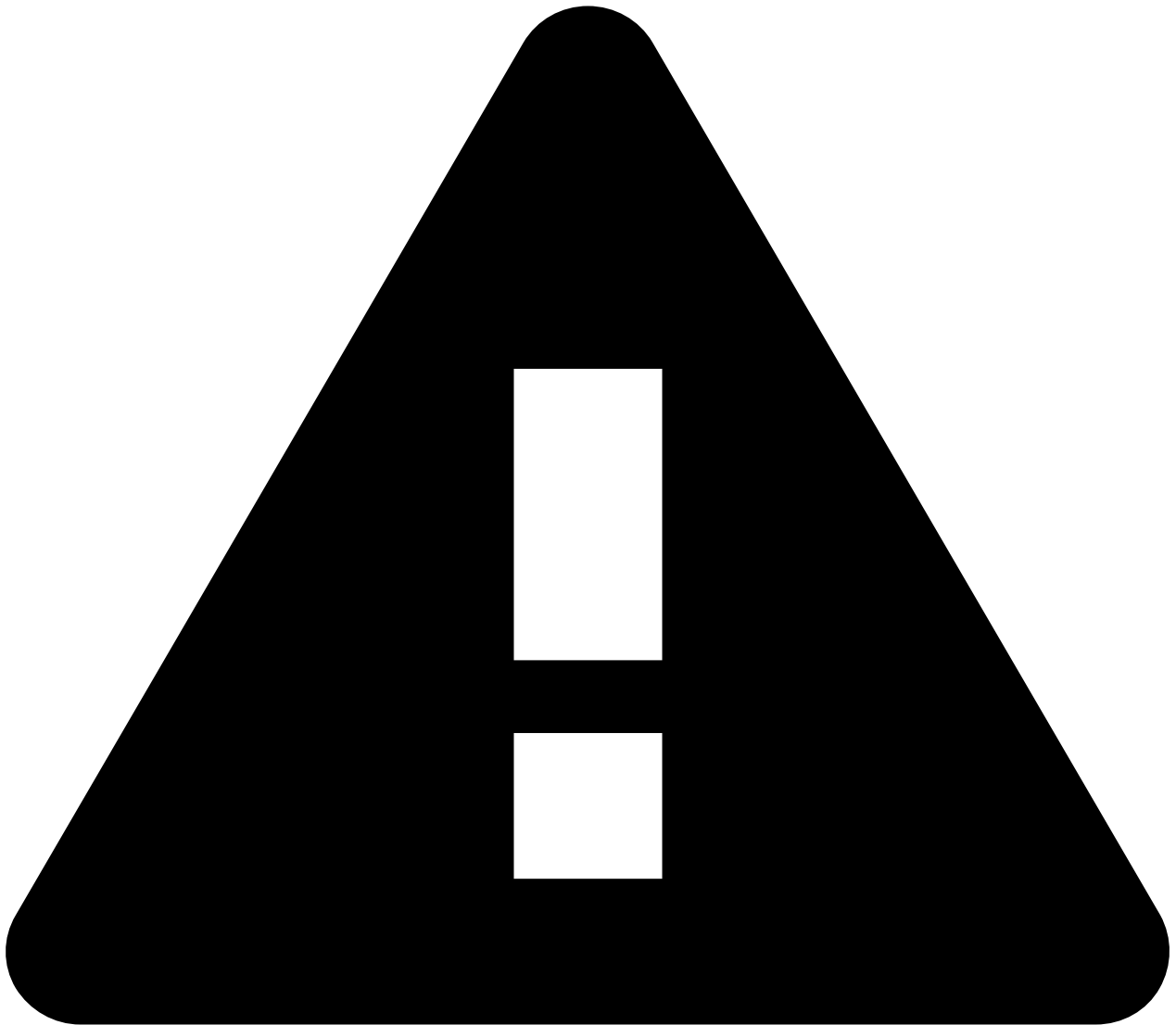
Visit the [Data Files](#) page for more details on managing data files.



tip

STIG Findings and severity-specific STIG queries can be run using the latest uploaded STIG Security Policy file, without waiting for the next snapshot. Use the selector in the verification details view to choose Snapshot or Snapshot with Latest Data Files.

See [Data Files for STIG Security Policies](#) for more details



[Important]

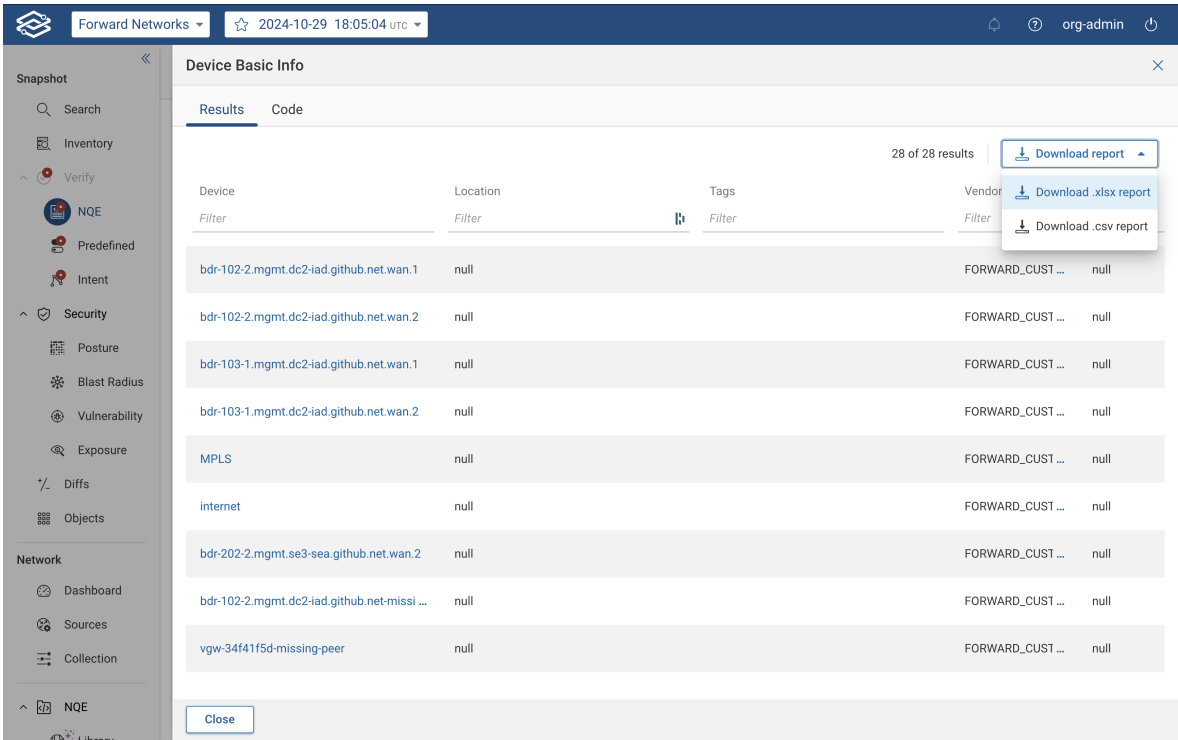
The verification status shown in the main Verify table always reflects the snapshot only. Previewing with latest data files does not modify snapshot-driven results.

# Downloading a Report🔗🔗

The **NQE Verifications** page allows users to export query results in various formats for analysis or compliance purposes. Available export formats include **CSV**, **XLSX**, and **CKLB**.

## Downloading a CSV and XLSX Report🔗🔗

1. Open the **NQE Verifications** page and locate the desired query and click the **View Details** icon.
2. In the results pane, click on **Download report** at the top-right corner.
3. Select one of the following options from the dropdown:
  - **Download .xlsx report**
  - **Download .csv report**



## CKLB Export🔗🔗

CKLB Export generates STIG compliance reports in a DoD-compliant [CKLB format](#). This export option is available only for queries that include all the required columns found in any of the **STIG Findings** queries (e.g., STIG Findings - Cat I, STIG Findings - Cat II, or



STIG Findings - Cat III). While additional columns can be added to the query, none of the required columns can be removed.

### To download a STIG report:

1. Open the **NQE Verifications** page and locate any of the **STIG Findings** queries.
2. In the results pane, click on **Download report** at the top-right corner.
3. Select **Download .ckl report (STIG)** from the dropdown.

The screenshot shows the 'STIG Findings' window in a software application. The window has a dark blue header with 'Forward Networks' and a date/time filter '2024-10-29 18:05:04 UTC'. On the left is a sidebar with navigation options: Snapshot, Search, Inventory, Verify, NQE, Predefined, Intent, Security, Posture, Blast Radius, Vulnerability, Exposure, Diffs, Objects, Network, Dashboard, Sources, Collection, and NQE. The main area is titled 'STIG Findings' and has tabs for 'Results' and 'Code'. Below the tabs is a table with columns: violation, Device, Vulnerability ID, Rule ID, Rule Version, Intent, and a 'Download report' button. The table shows 500 of 1,192 results. A dropdown menu is open from the 'Download report' button, showing options: 'Download .xlsx report', 'Download .csv report', and 'Download .ckl report (STIG)'. The table rows show 'Fail' status for various devices and rules.

| violation | Device   | Vulnerability ID | Rule ID               | Rule Version   | Intent                                                                                                                          |
|-----------|----------|------------------|-----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| Fail      | bigip3k1 | V-217383         | SV-217383r879525_rule | F5BI-DM-000019 | The BIG-IP appliance must automatically ...                                                                                     |
| Fail      | bigip3k1 | V-217384         | SV-217384r879526_rule | F5BI-DM-000021 | The BIG-IP appliance must automatically ...                                                                                     |
| Fail      | bigip3k1 | V-217385         | SV-217385r879527_rule | F5BI-DM-000023 | The BIG-IP appliance must automatically ...                                                                                     |
| Fail      | bigip3k1 | V-217386         | SV-217386r879528_rule | F5BI-DM-000025 | The BIG-IP appliance must automatically ...                                                                                     |
| Fail      | bigip3k1 | V-217387         | SV-217387r879530_rule | F5BI-DM-000027 | The BIG-IP appliance must be configured for each administrator and authorization the privilege level in accordance with app ... |

## Additional Options

An **Additional Options** icon is available for Verifications with defined query parameters:

- **Edit Parameters** - Selecting this option opens the edit parameter settings window for editing.
- **Duplicate** - Selecting this option will duplicate the selected Verification and make a copy available for use.

Forward Networks

2023-12-11 10:41:05 GMT-08:00

admin

Snapshot

Search

Verify

NQE

Predefined

Intent

Security

Posture

Blast Radius

Vulnerability

Exposure

NQE Verification

Directories

Search

Add NQE queries

Full report

All Verifications

Forward Library

Cloud

Devices

Config Search

Interfaces

L3

BGP

0 selected

Activate

Deactivate

Download result reports

2 Verifications

0 hidden columns

| Active                   | Name                                | Intent                              | Last committed           | Status       | Run time | Priority |         |
|--------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------|----------|----------|---------|
| Filter                   | Filter                              | Filter                              | Filter                   | Filter       | Filter   | Filter   |         |
| <input type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | Config Search-interface  | 2 months ago | failed   | <1s      | not set |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | Config Search-interfaces | 2 months ago | failed   | <1s      | not set |

Edit parameters

Duplicate